Certifications Essentials

AWS Regions, Availability Zones, and Edge Locations

- Region: Completely separate geographic area hosting AWS data centers. Provides the greatest possible fault tolerance and stability, and allows you to deploy workloads around the globe
- Availability Zone (AZ): An isolated, independent location within a specific Region with their own redundant power, networking ,and low-latency connectivity. Physically separated to reduce impacts of disasters
    - Different AZs may map to different AZ IDs on different accounts
        - One Account: us-west-2b (AZ) -> usw2-az1 (AZ ID)
        - Another Account: us-west-2b (AZ) -> usw2-az3 (AZ ID)
            - AZ IDs are consistent across accounts
- Edge Location: AWS data centers meant to help you deliver services with the lowest latency possible. Closer to end users compared to Regions and AZs. Leverages “Points of Presence” and Regional edge caches

Shared Responsibility Model and Well-Architected Framework

- Customer Responsibility (Security in the cloud)
    - Encryption of Customer Data
    - Backup up of Customer Data
    - Controlling access and authorization
    - Managing operating systems, network configurations, and firewall controls
- AWS Responsibility (Security of the cloud)
    - Software behind the AWS Services
    - Physical security and redundancy of infrastructure
    - Underlying hardware maintenance
    - Global Infrastructures (Regions, AZs, Edge Locations)
- Well-Architected Framework is a set of principles
    - Operational Excellence (Does your architecture work and will it continue to?)
        - Support Development
        - Run workloads effectively
        - Gain insights into operations
        - Continuously improve supporting processes and procedures
    - Security (Does this system work only as intended?)
        - Protect data
        - Protect systems
        - Protect assets
    - Cost Optimization (Spend only what you have to)
        - Operate at the lowest price point
    - Reliability (Will this system work consistently and recover quickly?)

        - Perform correctly and consistently
        - Be easily tested
    - Performance Efficiency (Remove bottlenecks and reduce waste)
        - Use correct computing resources
        - Maintains efficiency with scaling demands
    - Sustainability (Minimize environmental impacts from workload)
        - Reducing energy consumption
        - Increasing efficiency across all components of a workload
        - Maximize the benefits from provisioned resources
        - Minimizing total resources required
- Well-Architected Tool
    - Service providing consistent process for measuring your architecture
    - Assists with documenting the decisions and gives recommendations
- High Availability
    - Designed to remain operational for long periods of times
    - Failures result in minimal downtime
    - Redundancy and failover mechanisms allow quick recover
    - Minimizes downtime through redundancy/failover
- Fault Tolerance
    - Systems continue to operate normally even if one or more component fail
    - Aims for little/no interruption at all
- Recovery Time Objective (RTO)
    - Maximum acceptable amount of time to restore a system
- Recovery Point Objective (RPO)
    - Maximum acceptable amount of data loss measured in time

Core Services

IAM Overview

- IAM: Unique authentication database that is logically isolated to one AWS account
    - Create users and grant permissions to those users
    - Create groups and roles
    - Control access to AWS resources and services
- Root User Account:
    - Email address (unique) you used to sign up for AWS, has full administrative access
    - Use MFA immediately and avoid using for normal tasks and do not create access keys

IAM Users and Groups

- IAM User: Entity within IAM meant to represent a human or a dedicated service account

- IAM Group: Collections of IAM users to simplify permission management. Users can belong to multiple IAM groups at one time
- Long term Credentials: Authentication via username and password, or by static IAM Access Keys
- Policy: Object in AWS that, when associated with an identity or resource, defines their permissions
    - Identity-based
        - Attached to IAM identities (User, group, or role)
        - Written and stored as JSON
            - Managed Policies: Attachable, standalone policies that are reusable
                - AWS Managed Policies: Created and managed by AWS, usable by everyone
                - Customer Managed Policies: You create, manage, and reuse however you want
            - Inline Policy: Added directly to a single IAM identity for specific use case
                - 1 to 1 relationship, not reusable
    - Resource-based
        - JSON policy documents that get attached to AWS resources like Bucket Policy or KMS Key Policy
        - Grant permissions to a specified IAM principal for the resource
        - All resource-based policies are inline policies
        - Grant cross-account by specifying another account as the principal
    - Permission boundaries
    - Organization service control policies
    - Access Control Lists
    - Session Policies
- IAM Policy

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "EnforceMFAForSensitiveS3Actions2026",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::123456789012:user/ComplianceOfficer"
            },
            "Action": [
                "s3:PutObject",
                "s3:DeleteObject"
            ],
            "Resource": [
                "arn:aws:s3:::financial-records-2026/*",
                "arn:aws:s3:::audit-logs-2026/*"
            ],
            "Condition": {
                "Bool": {
                    "aws:MultiFactorAuthPresent": "true"
                }
            }
        }
    ]
}
```

- Version: Required (Currently always "2012-10-17")
- Statement: Required, list containing permission statements
    - Sid: Optional, identifier for your statements in policy
    - Effect: Required, allowing or denying access
    - Principal: Depends, IAM identity that you are applying this policy to using ARN
    - Action: Depends, Which API calls (actions) are you allowing or denying dependent on resources
    - Resource: Depends, list of AWS resources that you are applying the actions to
    - Condition: Optional, special conditions that you can set to further customize policies

- Access Keys
    - Long-term credentials for IAM users and the root user account
    - Used to sign programmatic requests via CLI or SDK
    - Composed of Access Key ID and Secret Access Key
        - Access Key ID is like a username
        - Secret Access Key is like a password
    - Secret Access Key is only viewable at time of creation
- IAM Credential Reports
    - Report that you generate and download that lists

        - All users in the account
        - Status of passwords
        - Status of access keys
        - MFA Status
        - Useful for auditing and compliance, data updates every 4 hours

IAM Roles

- Another type of IAM identity that you create in your account that has specific permissions
- No longer-term credentials used, temporary and rolling
- Useful for delegating access to many users, apps, or services
- Control permissions and access via permissions policies and trust policies
- Types
    - Service-linked roles: Allow AWS services to access AWS resources securely
    - Instance Profiles: Attached to EC2 instances to allow use of IAM Role
    - Federated Identities: Allows external identities to assume roles
- Common Use Cases
    - Grant access to Lambda functions to allow them to access resources
    - Grant access to EC2 instances to allow them to interact with services
    - Set up a role to allow cross-account access from another AWS account for third-party vendors
- AWS Security Token Service (STS)
    - Enables you to request short-term, temporary credentials for users, lasting 15 minutes to 12 hours
    - Commonly used with SAML federation and OIDC (OpenID) federation
- SessionToken on top of AccessKeyId and SecretAccessKey is required
- Trust Policy: Defines which principals are trusted to assume the role
    - Principals can be IAM users, IAM roles, AWS accounts, AWS Services
    - Required for a role to be usable

```json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::123456789012:root"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId": "unique-external-id-12345"
        }
      }
    }
  ]
}
```

- Principal: Can assume the role
- EC2 Instance Profiles
  - Used to pass an IAM role to EC2 instance
  - Creating a role for EC2 in console results in automatic instance profile creation
  - Required for leveraging IAM role credentials with EC2 instances
    - Hypervisor requires instance profiles to pass IAM role credentials to EC2 instances
    - Instance Profile contains the role
- IAM Roles should be used whenever possible since they use temporary credentials which are more secure
- Trust Policy vs IAM Policy
  - Trust Policy determines who can use the role
  - IAM Policy determines what the role can do
  - Both are necessary, or else the role will be useless

AWS CloudShell

- Browser-based shell that allows you to interact with AWS resources
- Creates a pre-authenticated session using console credentials
- Pre-installed software and tools

VPC Overview and CIDRs

- Logically isolated part of AWS Cloud where you can define your own network, including virtual networks, IP address ranges, subnets, route tables, and network gateways

- Like your very own virtual data center network in the cloud
- Types
    - Default VPC: Automatically created with AWS account
        - Reachable and publicly resolvable on the internet, not recommended to use due to security consequences
        - Comes with DHCP Option Sets, Main (default) route table with a primary local route, and Main (default) NACL with 2 simple rules
    - Custom: Created by you with custom CIDR block range
- Components
    - CIDR Blocks
    - Subnets
    - Route Tables
    - Security Groups
    - NACLs
    - Network Gateways
- You select an IP CIDR block to use for networking resources in VPC
- Private IPv4 CIDR blocks are required from ranges /16 to /28
- IPv6 CIDR blocks are optional from /44 to /60
- CIDRs must be from the RFC 1918 range (private IPs): 10.0.0.0, 172.16.0.0, 192.168.0.0
- VPCs are Regional
    - Soft limit of 5 VPCs per Region
- IPv4 CIDR Example
    - 10.0.0.0/16
        - IPv4 range is made up of 32 total bits
        - Every 8 bits is separated by dots (8bit.8bit.8bit.8bit./16)
        - /16 represents the amount of bits for the subnet mask, translates to 255.255.0.0, meaning that the first 2 octets (10.0) are used for network addresses, last 2 octets which are 16 remaining octets (0.0) are the total available IP Addresses
        - 2^(# of remaining bits)
        - 32 total bits - 16 used bits = 2^16 = 65,536 IPs

VPCs: Subnets, Routing, NACLs, and Security Groups

- VPC Internet Gateways: Allows you to communicate between VPC and internet
    - Supports IPv4 and IPv6
    - Automatically scales and offers high availability
    - Enables public subnet resources to connect to internet
    - Gives you a target in your VPC for internet-routable traffic
    - Created separately from VPC, only attachable to 1 VPC
- VPC Subnets: Range of IP addresses within VPC for hosting resources
    - Bound to a single Availability Zones
    - Supports IPv4, IPv6, dual stack (both at the same time)
    - Types: Public, Private, VPN-only, Isolated (No connectivity)

    - AWS reserves 5 IP addresses per subnet
        - Example: We have a /28 subnet CIDR
            - Total bits in IPv4 is 32 bits, bits used in this network is 28, 32-28 = 4, 2^4 = 16
            - Since 5 addresses are reserved, only 11 usable IP addresses
        - Example: VPC CIDR: 10.0.0.0/16, Subnet CIDR: 10.0.0.0/24
            - Network Address: 10.0.0.0
            - VPC Router: 10.0.0.1
            - VPC DNS Server: 10.0.0.2
            - Future Use: 10.0.0.3
            - Broadcast Address (AWS Reserved): 10.0.0.255
    - Public Subnets: Direct route to Internet via Internet Gateway
    - Private Subnets: No direct route to internet, requires NAT device to access internet
    - Since Subnets are bound to a single AZ, and we want to follow the high availability and redundancy according to the well architected framework, we should use multiple subnets across multiple AZs
    - Tier: Logically isolated section in the network
        - Having multiple tiers allows a more secure network by using “security in layers” approach
        - You can restrict the layers that are able to communicate to each other
        - Multi-tiered (2 or more layers of subnets) VPC designs are best practice
        - Most common design is to have Web (Public Facing), App (Only allows traffic from web tier), and Database tiers (Only allows traffic from app tier)
    - Subnets can communicate across AZs automatically, but there is a cost associated
- VPC Route Tables: Contains rules (routes), that tell your network traffic where to go
    - Prioritizes longest prefix match (Strictest rule prioritized if overlap)
    - Types:
        - Main Route Table: Automatically comes with VPC and acts as the default table for leftover, unassociated subnets
        - Custom Route Table: You fully define and associate with subnets
    - Concepts and Terms
        - Destination: Range of IP Address where you want to direct your traffic towards (Ex: 192.168.0.0/24)
        - Target: The different gateway, network interfaces or other connections where the destination traffic should go (Ex: Internet Gateway)
            - Destination is reached through the target
        - Local Route: Every route table has a local route applied for any VPC-bound traffic
            - Ensures that any resource inside your VPC can talk to any other resource in the same VPC by default.
        - Association: You associate subnet with a route table to apply chosen rules for any network traffic in the subnet

    - Generally best practice to have 1:1 Relationship for subnets and route tables
- Network Access Control Lists (NACLs): Allows or denies specific inbound or outbound traffic at the subnet level
    - Stateless firewall to control traffic at the subnet level
        - Stateless: Explicitly define both inbound and outbound traffic rules
        - Requires both inbound and outbound rules
    - Assign one NACL per subnet
    - Newly created NACLs will deny all traffic by default
    - List of ascending numbered, prioritized rules where the first match wins
    - Doesn’t work with: Amazon Domain Name Services (DNS), Amazon Dynamic Host Configuration Protocol (DHCP), Amazon EC2 instance metadata, Reserved IP addresses used by the default VPC router (These bypass NACL rules)
    - Ephemeral Ports: Short-lived transport protocol ports that operating systems allocate for client-side transmissions when they connect to a server
        - Vary by operating system
        - Example: Inbound HTTP connections use port 80, but outbound will be between 1024-65535
- Security Groups: Controls traffic that is allowed to reach and leave the resources it is associated with
    - Stateful virtual firewalls attached at the EC2 and network interface level (resource level)
    - Only define allowed traffic, no deny rules ever
    - Implicit deny is in place
    - Components
        - Protocol (TCP, UDP, etc.)
        - Port Range (22, 443, etc.)
        - Source or Destination (Which source or IP ranges are allowed in and out)
    - Can reference other security group IDs within your security group rules
- DHCP Option Sets: Group of network settings used by resources in VPC
    - Allow you to control aspects of network configuration in VPC
        - DNS Servers
        - Domain names
        - NTP Servers
        - Whether you want DNS resolution turned on or off in VPC
    - Can associate option set with multiple VPC
    - Each VPC can have only 1 associated DHCP option set
    - Cannot modify DHCP option set once it is created (Need to create a new one and associate it with the VPC)

VPC Peering, Network Gateways, Endpoints, and AWS PrivateLink

- VPC Peering: Enable secure and direct communication between VPCs, traffic remains within AWS network infrastructure
    - No single point of failure for connection
    - Connect Cross-Account, same account, and even cross-region

    - Requestor VPC: VPC that sends the peering request
    - Acceptor VPC
    - Need to update VPC router tables to correctly route traffic between VPCs and configure NACLs and security groups
    - Can enable VPC to resolve public IPv4 DNS hostnames to private IPv4 addresses when queried from instances in the peer VPC
        - Both VPCs must be enabled for DNS hostnames and DNS resolution
    - Peered VPCS cannot have any overlapping IP CIDRs
        - The mask is how many bits are “locked” from left to right
        - Ex: 10.0.0.0/16 -> 10.0.0.0 to 10.0.255.255
    - Peering does not allow for transitive routing
        - If VPC A is peered to VPC B, and VPC A is also peered to VPC C, VPC B cannot talk to VPC C through VPC A
    - When VPCs are peered in the same region, you can reference peered VPC Security IDs as needed
    - Use Cases
        - Centralized Shared Services
        - Multi-Region Internal Application
        - Cross-account VPC Integration
- Public NAT Gateway
    - Network Address Translation (NAT): Service that operates on a router or edge platform to connect private networks to public networks, with NAT, an organization needs one IP address or one limited public IP address to represent an entire group of devices as they connect outside their network
    - AWS NAT Gateway: Can use this so instances in a private subnet can connect to services outside VPC but external services cannot initiate connection with those instances
        - Automatically scales, deployed to a single AZ (redundant), leverages an Elastic IP address
            - Elastic IP Address (EIP): Static IPv4 address designed for dynamic cloud computing, are allocated to your AWS account and do not change
        - Should deploy a NAT Gateway in multiple AZs for high availability
        - You do not assign security groups to NAT gateways, can control access via NACLs but those are at the subnet level
    - Private: Allows resources in private subnets to connect to the internet, peered VPCs, and on-prem networks
    - Secure: Private resources cannot receive unsolicited connection request from outside the network
    - NAT Instance: Outdated and should be avoided, provides network address translation via an EC2 instance that you own and manage, lots of overhead
        - Requires disable Source/Destination Check option on EC2
    - Must deploy a NAT device (Gateway) within a public subnet to allow internet access

- Transit VPCs: Can achieve Transit VPC by using a VPN solution
    - Setup: Create EC2 Router/VPN inside this Transit VPC, use Virtual Private Gateways in your VPC to connect to those routers which uses an IPSec VPN connection tunnel
        - Virtual Private Gateway (VGW): VPN concentrator on the Amazon side of a site-to-site VPN connection
            - Attach them to a VPC that needs access to the VPN connection
- VPC Endpoints and AWS PrivateLink
    - AWS PrivateLink: Privately connect your VPC to services as if they were in your VPC
        - Every AWS service is going to leverage the public endpoint by default
        - Removes needs of IGWs, NAT Gateways, VPNS, Direct Connects
        - Create endpoints within VPC to control and secure traffic
        - Capable of hosting your own private services by setting up a service provider and a service consumer
            - Service Provider: VPC that's hosting the application you want to share out, creates a network load balancer to front the shared, private service
            - Service Consumer: Interface VPC endpoints (shared ENI) is created within the consumer VPC connecting to the provider NLB
        - AWS PrivateLink powers VPC Interface Endpoints
        - Good for peering to tens, hundreds, or thousands of VPCs
- VPC Gateway Endpoints: Provides connectivity to S3 and DynamoDB without requiring an internet gateway or NAT device for VPC, are free
    - Does not leverage AWS PrivateLink
    - Requires updates to route tables
    - Easy to use and reference via a Managed Prefix List for routes
    - Do not require any security group updates
- VPC Interface Endpoints:
    - Deploys an Elastic Network Interface into chosen VPC subnets
    - Requires management of an attached security group
    - Supports more services than gateway endpoints, but costs money for each ENI
    - Traffic is sent to AWS services via private endpoints, which require use of regional and zonal DNS names
        - Regional DNS Name: vpce-078bad00d40f22a11.logs.us-east-1.vcpe.amazonaws.com
            - VPCEndpointID.ServiceID.Region.VPCEndpoint
        - Zonal DNS Name: vpce-078bad00d40f22a11-us-east-1b.logs.us-east-1.vcpe.amazonaws.com
            - VPCEndpointID-AZName.ServiceID.Region.VPCEndpoint
    - Private DNS: Enables you to make requests to a service using public endpoint DNS name, while using private connectivity via the interface VPC endpoint

    - Must deploy a VPC Interface Endpoint into a private subnet, which deploys an ENI into the subnet, which uses PrivateLink
    - S3 can leverage both types of endpoints
    - Offers much more control over Gateway Endpoints, but Gateway is free and easy to use, so depends on situation

EC2 Overview

- EC2 and AMIs
    - IaaS (Infrastructure as a service)
    - Can grow or shrink the instance as you need it
    - Basics
        - Compute (Number of VCPUs)
        - Memory (RAM)
        - Network (Bandwidth)
        - Storage
    - Amazon EC2 Key Pairs
        - Private and Public Key
            - Public Key is stored on EC2
            - Private Key is for connecting
        - Used for SSH into instances and decrypting admin password for window instances used to RDP
    - Amazon Machine Image (AMI)
        - Public: Anyone with AWS account can explore and launch
        - Private: Create your own, choose to keep private or share with other AWS accounts
        - Prebaking: Embedding a significant portion of your application artifacts so it's ready to go
- Sizes and Instance Types
    - What to consider:
        - Class and Generation
            - Example: t4g.large
                - Family(t): General Purpose Burstable
                - Generation (4g)
                - Size (large)
            - Instance Types:
                - General Purpose: Balanced, Primarily (M & T (Burstable) Type)
                - Compute Optimized: High Performing Processes (C Type)
                - Memory Optimized: Processing large data sets, high memory per CPU ratio (R, X, Z Types)
                - Accelerated Computing: Hardware acceleration, useful for video transcoding and graphics rendering (P, G instances)
                - Storage Optimized: Workloads that need high, sequential read and write access to local storage (I instance Family)

            - HPC Optimized: High Performance Computing, Genomic simulations, ML (HPC instance Family)
        - Instance Size
- EC2 User Data
    - Passed into an instance to perform automated configuration tasks and run scripts right after instance starts
        - Known as Bootstrapping
    - Formats
        - Simple plain text
        - Base 64 Encoded
            - Good for long scripts
    - Good for configuring instance settings, joining to a domain, and bootstrapping applications
- EC2 Hibernate
    - Suspend to Disk
    - Saves content in RAM to EBS volume
    - Useful for quick recovery failures
    - Starting after Hibernation
        - EBS root volume is restored to previous state
        - RAM content are reloaded
        - Processes previously running are resumed
        - Previously attached data volumes are reattached

EC2 and EBS

- File Storage: EFS and FSx
- Block Storage: EBS and Instance Store (Temporary)
- EBS: Volume created and attached at instance launch is called root data volume
    - Are network attached drives that act like locally attached drives (Can persist after instance shutdown, but default behavior is to delete volume when instance is terminated)
    - Volumes are bound to one AZ (EC2s are also bound to one AZ too)
    - Types:
        - General Purpose SSD (gp2 and gp3): Boot disks and general applications
        - Provisioned IOPS SSD (io1 and io2): Latency sensitive applications, use this if you really need performance that gp does not provide (gp provides 16000 IOPS while this provides 64000 IOPS)
            - io1 is most expensive, io2 is more durable
        - Provisioned IOPS SSD (io2 Block Express): Storage Area Network (SAN) in the cloud, highest performance, sub-millisecond latency: Largest, most critical high performance operations (256000 IOPS)
        - Throughput Optimized HDD (st1): Suitable for big data, data warehouses, ETL, can't be a boot volume
        - Cold HDD (sc1): Less frequently accessed data, can't be a boot volume

        - EBS Multi-Attached Volume: Can attach to up to 16 instances at once (only work in same AZ), must use cluster-based file system (GlusterFS for example)
    - IOPs: Read and write speeds (good for fast data)
    - Throughput: Bits read or written per second (good for big data)
- Can encrypt both boot and data volumes
- EBS Snapshots
    - Incremental
    - Captures everything at time of taking snapshot, can be shared across regions
    - Stored in S3 (You don't't' have access to these)
    - Can Archive snapshots
    - Recycle Bin is for recovering accidentally deleted snapshots
- EBS Fast Snapshot Restore: Can create volume from a snapshot quickly (Only 16 TiB or less)
- EC2 Instance Store: Temporary Block Storage
    - Located on disks physically attached to host of EC2
    - Only attached at EC2 launch and not after
    - Persists with reboots but is erased with stopping, hibernating, and terminating
    - Size of instance store depends on instance size and type, you can't control it

EC2 Security Features

- Bastion Host: Servers running at the edge of a network allowing secured and controlled access into private network, also called jump server/box
    - Typically deployed into public subnet and leverages SSH (Port 22)
    - Their firewalls can be leveraged to perform port forwarding
    - Can force authentication and security requirements
    - Can use port forwarding (Ex: ssh -L 8080:rds_instance:5432 user@bastion)
    - Systems Manager Session Manager is preferred over bastion hosts
- Connecting to EC2 via Instance Manager
    - Only some Linux OS allow
    - Requires public IPv4 address
    - Uses SSH from an Amazon IP
- Connecting to Windows EC2 via RDP
    - Uses Key Pair to decrypt admin password
    - You use this password to connect via RDP
- Connecting to EC2 via SSH
    - Requires Key Pair
- Session Manager: Provides an agent-based connection to managed EC2 instances, edge devices, and managed on-prem VMs
    - No SSH and RDP requirements
    - IAM policies control access
    - Log and audit connections via CloudTrail, S3, and CloudWatch
    - SSM Agent: Software that runs on instances and makes it possible for Systems Manager to update, manage, and configure these resources

- Requirements
    - Windows, Linux, or MacOS
    - IAM permissions
    - SSM Agent has to have network access to Systems Manager service, either via Internet or VPC interface endpoints
- Session Manager VPC Interface Endpoints
    - ssm.your-region.amazonaws.com
    - ssmmessages.your-region.amazonaws.com
    - ec2messages.your-region.amazonaws.com
- Can connect to EC2 via Session Manager via CLI (aws configure, then aws ssm start-session –target “i-123..”

- EC2 Metadata: Data about your EC2 Instance
  - Instance Metadata Service (IMDS)
    - IMDSv1: HTTP request/response
      - Example: curl http://169.254.169.254/latest/meta-data/instance-type
    - IMDSv2: More secure requiring headers, requires session token
      - Example: TOKEN=`curl -X PUT “http://169.254.169.254/latest/api/token” -H “X-aws-ec2-metadata-token-ttl-seconds: 21600”` && curl -H “X-aws-ec2-metadata-token: $TOKEN” http://169.254.169.254
      - Then: curl -H “X-aws-ec2-metadata-token: $TOKEN” http://169.254.169.254/latest/meta-data/instance-type
  - Can get user data, host name, security group information, instance IAM credentials
  - Can be set so only IMDSv2 can be used
  - Retrieving Metadata via IPv4: 169.254.169.254
  - Retrieving Metadata via IPv6: fd00:ec2::254
    - Requires subnet with IPv6, and EC2 instance must be AWS Nitro System instance
  - You can disable the metadata service if you want

EC2 Networking and Performance Scenarios

- Elastic Network Interface (ENI): Logical Networking component that is part of a VPC and represents a virtual network card for compute resources
  - Created and bound within a single Availability Zone (Makes sense since EC2s are bound to one AZ)
  - Can attach, detach, and reattach as needed
  - Attributes:
    - Primary IPv4 address or IPv6 from subnet CIDR
    - Secondary IPv4 address from subnet CIDR
    - One Elastic IP Address for each private IPv4 address
    - Security Groups with customized rules
    - One Ephemeral public IPv4 address

        - Ephemeral = temporary
        - When you assign a public IP, it takes it from Amazon's pool of public IPv4 addresses, when it is dissociated from instance, it goes back to the pool
- Elastic IP Addresses: Public, static IPv4 address that you can map to different compute
    - Can only be used within specific region
    - Allocate an address, then associate the address
    - Soft limit of 5 EIPs per region, per account
    - Public IPv4 address given to EC2 instances change after stops, so you can assign an Elastic IP Address (EIP) to the instance so it never changes
    - Avoid using these if possible, you should use DNS names that you can map to ephemeral IP addresses instead
- Dual-Home EC2 Instances: Multiple ENIs attached which live in different network bands
    - EC2 networking is split between 2 different subnet/tiers
    - 2 different subnets, 2 different security groups
- EC2 Placement Groups: Optional grouping of EC2 instances to influence their placement
    - Cluster: Puts instances closely together in the same AZ (clusters them together), low latency network performance, good for high-performance computing (HPC)
        - Not all instances types support this
        - AWS recommends homogenous instances (instances of the same type like m5.large) in cluster placement groups
    - Spread: Place small group of instances across different underlying hardware, meant to reduce common hardware failures, recommended for critical applications for uptime (all different hardware, spread instances out)
    - Partition: Spread instances across logical partitions, ensures groups do not share underlying hardware, each partition has its own set of racks
        - Key difference between this and Spread: Used for distributed/replicated workloads, good for Hadoop, Cassandra, Kafka
    - You can add a stopped instance to existing placement group
    - Cannot merge placement groups
- AWS Outposts: Brings AWS data center directly to you, on prem
    - Allows you to have large variety of AWS services in your data center
    - 1U or 2U servers (Outposts Servers), 42U rack (Outposts Racks)
    - Allows you to create a hybrid cloud where you can leverage AWS services inside your own data center
    - Brings Management console, APIs, SDK into your data center
    - AWS can manage the infrastructure for you
- Enhanced Networking for EC2:
    - Networking Card Types
        - Elastic Network Interface (ENI): Day to day networking tasks
        - Enhanced Networking (EN): Single root I/O virtualization (SR-IOV) for high performance
            - SR-IOV: Allows single physical network device to present itself as multiple virtual devices to OS and hypervisor

            - Allows much higher I/O performance while minimizing CPU
            - Options depend on instance type
                - Elastic Network Adaptor (ENA): Supports up to 100Gbps, choose this if possible
                - Intel 82599 VF Interface: Supports up to 10 Gbps, typically used on older instances
        - Elastic Fabric Adaptor (EFA): Accelerates HPC and machine learning (ML) applications
            - Offers OS-Bypass, allows applications to communicate directly with network interface hardware
            - Provides lower, more consistent latency, and higher throughput
            - Only supports Linux currently

EC2 Price Optimizations

- Reserved instance: Commit to a 1 or 3 period commitment
    - No Upfront (Least Savings), Partial Upfront, All Upfront (Most Savings)
    - Zonal (AZ) or Regional Reserved Instance (More flexibility, less discount)
    - Convertible Option allows you to change instance type, family, OS, etc. (Less discount)
    - Perfect for long running, critical applications
    - You can buy and sell reserved instances on AWS Marketplace
- Capacity Reservations
    - Reserve compute capacity for EC2 in specific AZ for any duration of time
    - Charged On-Demand Pricing even if not used
- Savings Plan
    - Compute Savings Plan: Most flexible, automatically applies to EC2 instance, Lambda and Fargate is included
    - EC2 Savings Plan: Specific instance family in chosen AWS Region, more savings since more specific, you can only change OS, size and tenacy
    - SageMaker Savings Plan: Automatically applies to SageMaker instance, like a Compute Savings Plan but for SageMaker instances
    - 1 and 3 Year Commitment Periods
    - Basically the successor to Reserved instances, reserved instances being phased out
- Dedicated Hosts and Instances
    - Dedicated Hosts: Physical servers dedicated for your use (most expensive, most control)
        - Supports Bring Your Own License (BYOL)
    - Dedicated Instance: EC2 instance ran on dedicated hardware
        - Partially supports Bring Your Own License (BYOL)
- Spot instances: Uses spare EC2 capacity that is available for high discount
    - Goes by hourly price (called spot price) which is adjusted based on long term supply and demand
    - Can only use if capacity is available

    - Perfect for flexible applications (2 minute warning)
    - Spot Block: Runs 1-6 hours and better avoids interruptions, more expensive than spot instances
    - EC2 Spot Fleets: Used to launch, hundreds to thousands of EC2 instances
        - Can be multiple instance types, across multiple AZs, combination of On-Demand and Spot, Automatic replacement of Spot Instances
        - Allocation Strategy:
            - Price Capacity Optimized: Chooses pool available based on capacity, then lowest price (recommended)
            - Capacity Optimized: Select pools with best amount of capacity
            - Diversified: Spot instances get distributed across all spot capacity pools, better for availability requirements
            - Lowest Price: Highest risk of interruption
- AWS Compute Optimizer: Uses ML to recommend optimal AWS Compute resources for your workloads to help reduce cost and improve performance
    - EC2 Instances
    - EBS Volumes
    - ECS
    - Lambda
    - RDS

Network Storage and EFS

- Elastic File System (EFS): Managed Network File System (NFS) that can be mounted by many EC2 instances
    - Can work across multiple AZs
    - Much more expensive (pay per use) than EBS due to high availability and scalability (scales automatically)
    - Uses standardized NFSv4.1 protocol
    - Only compatible with Linux AMIs
    - Can encrypt at rest using KMS
    - Use VPC Security Groups for controlling access
    - Great for content management (can share across EC2) and simple web servers
- EFS Performance
    - Allows up to thousands of concurrent NFS connections
    - 20 Gbps of throughput (Very fast)
    - Scales to petabyte sizes
    - Performance Modes
        - General Purpose (Default): Recommended, better latency
        - Max I/O: Higher per operation latency than General Purpose (despite the name)
    - Throughput Modes
        - Elastic (Default): Recommended, automatic scaling
        - Provisioned: Useful if you know performance requirements of your workloads

            - Bursting: Useful for throughput to scale with amount of storage in EFS
- EFS Storage Classes
    - Standard: Most expensive, high speed SSD storage (very fast and very responsive)
    - Infrequent Access (EFS-IA): Combination of low cost and high performance
    - Archive: Use if you only need to access few times a year or less, cheapest, minimum storage of 90 days
    - Can use lifecycle policies to move files between classes
- File Systems Types
    - Regional: Spreads across AZ
    - One Zone: Non critical workloads
- Amazon FSx for Windows: Fully managed, native Microsoft Windows file system for moving Windows-based applications files
    - Supports Windows NTFS and Server Message Block (SMB) Protocols
    - Supports AD users, ACLs, groups, security policies
    - Supports Distributed File System (DFS) namespaces and replication
    - Very Fast, Can scale largely, backs up to S3 daily, multi-AZ capable
- Amazon FSx for Lustre: Fully managed, parallel, distributed file system optimized for compute intensive workloads
    - Geared towards Linux based OS
    - Useful for HPC and ML workloads
    - Very fast and performant
    - Can use S3 as a file system
- Amazon FSx for NetApp ONTAP: Fully managed, shared storage with the popular data access and manage capabilities of ONTAP
    - Works with majority of OS
    - Scalable, supports NFS, SMB, iSCSI, and NVMe over TCP
- Amazon FSx for OpenZFS: Fully managed, file storage service that makes it easy to move data to AWS from on-prem ZFS or other linux-based file systems
    - Usable by macOS, Windows, Linux
    - Supports NFS
    - Multi-AZ, Single-AZ (highly available HA), Single-AZ (non-highly available non-HA)
    - Backups stored on S3

S3 Overview

- Simple Storage Service (S3): Secure, Durable, Scalable (Unlimited amounts) object storage
    - Store and retrieve any amount of data from anywhere at a very low cost
    - Regional resources
    - Objects can go up to 5TB in size, of any type
    - Cannot run an OS or DB
- S3 Buckets: Containers for storing all objects within S3
    - All AWS accounts share S3 namespace (must be globally unique)

- Buckets themselves are regional (even though names are global)
- Standard storage class redundantly stores objects on multiple devices across minimum of 3 AZs in a region
- Read-After-Write: After a write or overwrite, any subsequent read request immediately receives the latest object
- Strong consistency: You can immediately perform listing of objects with all changes reflected
- HTTP 200 if upload is successful

- S3 Objects: Files (data) that you upload to S3 buckets
    - Each object has a Key (name) and Value (data)
    - Prefixes are meant to serve as human-friendly naming structures (like folders)
        - Need to include forward slashes
    - Object key is the full path, including any prefixes
    - Objects can have multiple versions and metadata
    - Key example: s3://my-bucket/dev/2026/my_json_file.json
    - S3 Multipart Upload: Allows users to upload large objects in smaller parts
        - If any object part fails, you can retry the upload without affecting other parts
        - Good for very large files or bad connectivity
- S3 Storage Classes
    - Frequently Accessed Objects
        - S3 Standard
        - Express One Zone: High performance, single zone for single digit milliseconds data access (Maybe HPC or ML workloads)
    - Infrequently Accessed Objects (Retrieval fee)
        - S3 Standard-IA: Stores objects redundantly across multiple AZs
        - S3 One Zone-IA: Stores objects redundantly in 1 AZ
    - Rarely Accessed Objects (There is minimum storage requirement)
        - S3 Glacier Instant Retrieval: Long term data that requires milliseconds retrieval
        - S3 Glacier Flexible Retrieval: Retrieval in minutes (not real time)
        - S3 Glacier Deep Archive: Up to 48 hours for retrieval
    - S3 Intelligent-Tiering: Optimize storage costs by automatically moving data to most cost-effective access tier, you pay for monitoring and auto-tiering
- S3 Versioning (Bucket level)
    - Unversioned (default)
    - Versioning-enabled: Once it's on, you can only suspend, cannot turn off
    - Versioning-suspended
    - S3 inserts a delete marker when versioning is enabled, which becomes the object version
        - You remove the delete marker to restore object
    - Objects present before versioning enablement has “null” version
    - Can be integrated with lifecycle rules, and be set up to require MFA

- S3 Object Lifecycles: Lifecycle configurations automate moving objects between different tiers
    - S3 Lifecycle Rules: Transition Actions and Expiration Actions
        - Works on any versions
        - Transition Rule: Move tiers
        - Expiration: Expires object after some time
- S3 bucket replication (Asynchronous operation, not immediate)
    - Versioning must be enabled on both buckets
    - Existing objects are not replicated automatically
    - Deletions are not replicated
    - Cross-Region replication or Same-Region replication, can be different accounts

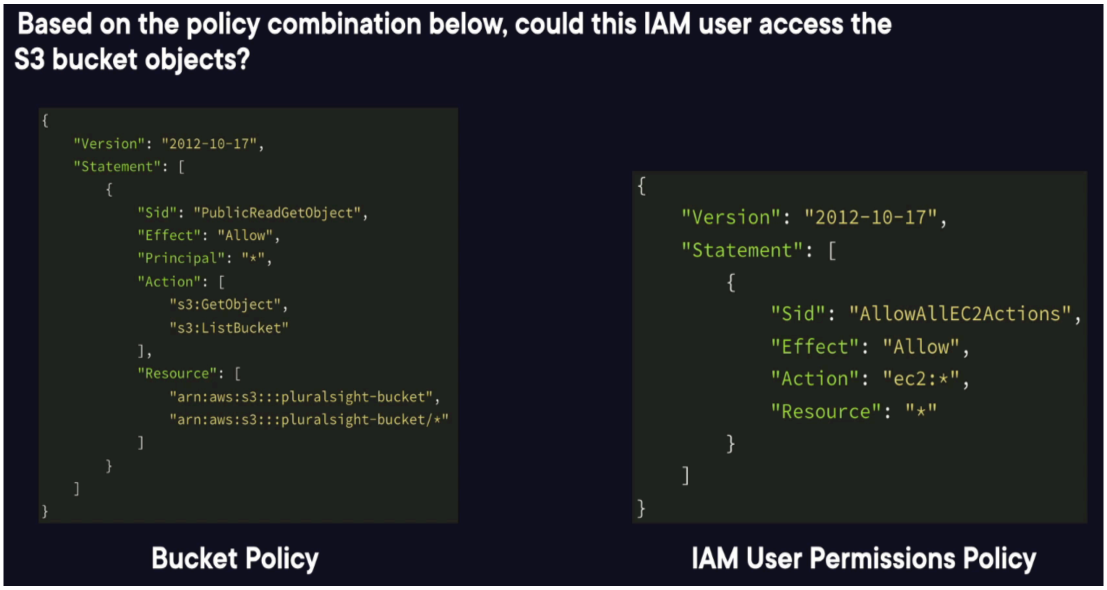
S3 Important Features

- S3 Batch Operations
    - Large-Scale Useful for performing single operation on large-scale set of objects
    - Jobs: Batch operations are ran by jobs, which use list of objects, actions to perform, and optional parameters
    - Tracks job progress, retries failed jobs, notifications, reports after completion
- S3 Select and S3 Glacier Select
    - No new customer access since Mid 2024
    - Best for small, simple queries, only works on single object at a time
    - S3 Select:Allows you to use SQL statements to filter content
        - Reduces amount of data that S3 transfers, which reduces cost and latency
    - S3 Glacier Select: Same as S3 Select but for Glacier
- S3 Storage Lens: Analytics feature used to gain organization-wide visibility into object storage and activity
    - Displays information on centralized dashboard, can export metrics
    - Can publish metrics and usage to CloudWatch
    - Free and Advanced Tiers
    - Metrics:
        - Summary
        - Cost-optimzation
        - Data-protection
        - Access-management
        - Event: Insights for S3 Event Notifications
            - S3 Event Notifications: Receive notifications when certain events happen in your bucket (Not immediate, can take few minutes)
                - Example Events: Object creation, removal, replicated, restored, lifecycle events, object tagging events
                - Destinations: SNS, SQS, Lambda, EventBridge Buses
        - Performance: Insights for S3 Transfer Acceleration
        - Activity (Advanced): Details about how storage is requested/uploaded
        - Status Code (Advanced): Detailed metrics on HTTP status codes

- S3 Transfer Acceleration: Bucket feature that speeds up data uploads and downloads
  - Uses customer-facing, globally distributed edge locations
  - Might not always result in performance gain (could be slower even)
- S3 requestor pays: Feature that allows you to offload costs of requests (transfer and storage) to originating caller
  - Requester must be authenticated via IAM
  - Per bucket
  - Requester must either include x-amz-request-payer header or RequestPayer parameter in REST request
- S3 Website Endpoints:
  - Public Read access is required
  - Ex: http://bucketName.s3-website-region.amazonaws.com
    - Dashes can be dots
- S3 Performance:
  - You can get better performance by spreading reads across different prefixes
    - This is because limits are put on requests per second per prefix, new prefixes circumvent this
  - S3 Byte-Range Fetch: Allows parallelize downloads by specifying byte ranges (specific portions of objects)
    - Better resiliency and higher throughput
    - This is like the counterpart/opposite for multipart upload

S3 Security

- S3 Access with Bucket Policies (Attached to Bucket)
  - Identity Based: Attached to IAM principals
  - Resource Based: Attached to AWS Resources
  - Bucket Policies are Resource Based
  - Can be used to allow public access, require encryption when upload, and give access to another AWS account
  - S3 buckets and objects are private by default
  - Block Public Access setting
    - Denys all public access to buckets, even with public access already set up (bucket and account level)
  - Tricky Scenario:

    - 

    - Yes, the IAM user can access the bucket, as long as there is an Allow in either place—and no Explicit Deny ("Effect": "Deny") anywhere
- Access Control List (ACL): List of grants identifying the grantee and permission granted
    - Used to grant basic read or write on other accounts
    - Uses S3-specific XML schema (hard to use)
    - Bucket ACL, Object ACL
    - AWS Account Canonical User ID: Unique, long-string identifier assigned to each AWS account, obfuscated form of AWS account ID
    - Canned ACLs: Premade ACLs
    - Recommended to not use ACLs (Use bucket policies if possible)
- Encrypting Data at Rest
    - All new buckets are encrypted at rest by default
    - Can enforce server-side encryption via bucket policy (x-amz-server-side-encryption header)
        - SSE-S3: S3 Managed Keys uses AES256 (Default)
            - No additional cost or performance impact
            - x-amz-server-side-encryption:AES256
        - SSE-KMS: Uses AWS KMS
            - x-amz-server-side-encryption:aws:kms
            - More control over encryption
            - KMS keys must be in same region as bucket
            - You must have decrypt permissions to read an object
                - Bucket Keys (Temporary key): Saves money, like a cached version of the key, saves on SSE-KMS cost
        - SSE-C: Customer Provided Keys
            - HTTPS is required
            - Key must be included in every request
                - x-amz-server-side-encryption-customer-algorithm (must be AES256)

                ○ x-amz-server-side-encryption-customer-key: The encryption key used for encryption and decryption
                ○ x-amz-server-side-encryption-customer-key-MD5: Used for integrity checks
            - No additional charges
        - Client-Side Encryption: You encrypt locally before uploading to S3
            - Can use S3 Encryption Client
- S3 Encryption in Transit
    - Should use TLS by using aws:SecureTransport (add to bucket policy)
- MFA Delete: Requires MFA for performing certain actions
    - Versioning is required
    - Bucket owner is only one who can enable this feature
    - MFA is required for permanent deletion of objects and suspending versioning
- S3 Access Logs
    - You specify a target bucket (where logs will be stored, same region)
        - You should use different buckets for storing logs to avoid infinite loops
        - Could take hours for logs to get delivered, best effort
- Presigned URLs
    - Can be used multiple times, up to specified expiry date and time
    - The URL uses the same credentials as the user generating, so if the user does not have sufficient permissions, the URL won't have access either
- S3 Access Points: Simplify data access for any AWS service or customer application that stores data in S3
    - Will appear as named network endpoints (DNS name) that are attached to endpoints
    - Used for performing S3 object operations with its own permissions
    - Can configure to only allow traffic from VPC, or configure a custom block public access setting for individual endpoints
- S3 Object Lambda: Add your own code to S3 GET, HEAD, and LIST requests to modify and process data as it is returned to an application by using AWS Lambda
    - Must create an S3 Object Lambda Access Point
    - Must create the Lambda function
    - Must create an S3 Access Point
    - Use cases: Removing PII, customized view of objects, convert data formats
    - Flow: User -> S3 Object Lambda Access Point -> Lambda Function -> S3 Access Point -> Bucket -> Lambda Function -> User
- S3 Object Lock: Locks the file from being deleted or modified for a fixed amount of time or indefinitely
    - Governance Mode: Protect objects against being deleted by most users
    - Compliance Mode: Projected object version can't be overwritten or deleted by anyone (including root)
    - Versioning must be enabled for both modes
    - S3 Object Lock Legal Holds: No retention periods and remain until removed
        - S3:PutObjectLegalHold permission

    - Glacier Vault Lock: Same thing as above but for Glacier

Route 53: Zones, Records, Policies, and Health Checks

- Global DNS with Route 53
    - DNS: Used to convert human-friendly domain names into IP address (IPv4 or IPv6)
        - http://mywebsite.com to http://44.224.44.122
    - Example Domain: www.google.com
        - .com is the top level domain (TLD)
        - google is the second-level domain (SLD)
        - www. Is the subdomain
    - Route 53: Highly available fully managed and scalable DNS web service
        - Fun Fact: Route 53 is the port for DNS traffic
            - Domain Registrations
            - DNS Routing
            - Health Checks
        - Authoritative DNS service:You control all updates of records
        - IPv4 Domain Registrar: Authority that can assign domain names, enforces uniqueness of domain names across internet
- Hosted Zones: Containers for records to specify how you route traffic for domains or subdomains
    - Public Hosted Zone: Routing traffic on public internet
    - Private Hosted Zone: Within VPC
    - Not free
- Route 53 Records
    - DNS Records: How and where you want your traffic routed
    - Domain or Subdomain Name: google.com
    - Record Type: A, CNAME, MX, NS, SOA
        - A: IPV4 Address
        - AAAA: IPv6 Address
        - NS: Name Servers
        - Alias: Special record that routes traffic to AWS resources
        - CNAME: Maps one DNS name to another domain, mywebsite.com -> website.com
            - You cannot have a CNAME for website.com (only one level)
    - Value: IP Address or AWS Resource Alias
    - Routing Policy
    - Time to Live (TTL)
- Routing Policy: How Route 53 responds to DNS queries
    - Simple Routing Policy (good for single resources and IPs)
        - Supports multiple values for the same record (chosen by random)
    - Weighted Routing Policy
        - Control percent of traffic routed to each traffic
        - Records must have same name and type

        - Weights do not have to add to 100 (relative weights)
    - Failover Routing Policy (Active-passive routing)
        - Routes traffic to a healthy resource if resource becomes unhealthy
        - Uses health checks
    - Multivalue Answer Routing Policy
        - Route 53 returns multiple values for DNS queries
        - Uses health checks to return only healthy resources
    - Geolocation Routing Policy
        - Traffic is sent based on genographic location of users (must have default)
    - Latency-based Routing Policy
        - Routes traffic based on lowest network latency (different regions)
    - Geoproximity Routing Policy
        - Policy that lets Route 53 to route traffic based on geographic location of users and resources
        - Bias: Expands or shrinks size of geographic region
- Health Checking Route 53 Resources
    - Can set health checks on individual record sets
    - Unhealthy records are automatically removed until healthy
    - Monitoring an endpoint
        - Route 53 uses global health checkers to monitor (can choose regions and intervals)
        - HTTP 2xx or 3xx status code returns
        - Make sure firewall rules allow health checkers
        - Only available for publicly resolvable records
    - Monitoring other health checks (Calculated health checks)
        - Combining multiple health checks into single one (up to 256)
        - Can use AND, OR, or NOT logic
        - Useful for multi-layered applications where the app should stay up as long as one endpoint is healthy
    - Monitoring CloudWatch alarms
        - Create CloudWatch metric and associate CloudWatch Alarm
            - ALARM state when triggered
            - Perfect for using within private hosted zone where no public access to records

Route 53: Resolvers

- Hybrid DNS: Resolves for both on-premises resources and cloud-based resources
- Route 53 Resolver (AmazonProvidedDNS): DNS resolver that responds recursively to DNS queries
    - VPCs connect to Route 53 Resolvers at a VPC+2 IP address and is available by default
    - VPC+2 IP address connects to a Route 53 Resolver within AZ
    - Works for Public records, Amazon VPC specific DNS names, Route 53 private hosted zones

- Route 53 Resolver Endpoints require a private connection (VPN or Direct Connection)
    - Inbound Resolver Endpoint: Traffic coming into VPC
    - Outbound Resolver Endpoint: Outbound requests coming out of VPC
    - Resolver Rule: Forwarding rule for where you want to send inbound or outbound traffic
        - Applied to VPCs and can be shared using AWS Resource Access Manager (RAM)
- Example Flow:
    1. On-prem server wants to resolve a website, so it sends DNS query to on-prem DNS resolver
    2. On-prem DNS resolver forwards query to inbound resolver endpoint in AWS VPC
    3. DNS query sent to VPC over private connection
    4. Inbound endpoint queries private hosted zone information for the resource via Route 53 Resolver
    5. Traffic is forwarded to correct private IP using private hybrid DNS
- Route 53 Resolver DNS Firewall: Allows you to control access to sites and block DNS level threats for DNs queries going out from VPC through Route 53 Resolver
    - Define domain name filter rules, can specify lists of domain names to allow or block
    - Only for DNS traffic
    - Same concept as Pi-Hole DNS Sinkhole

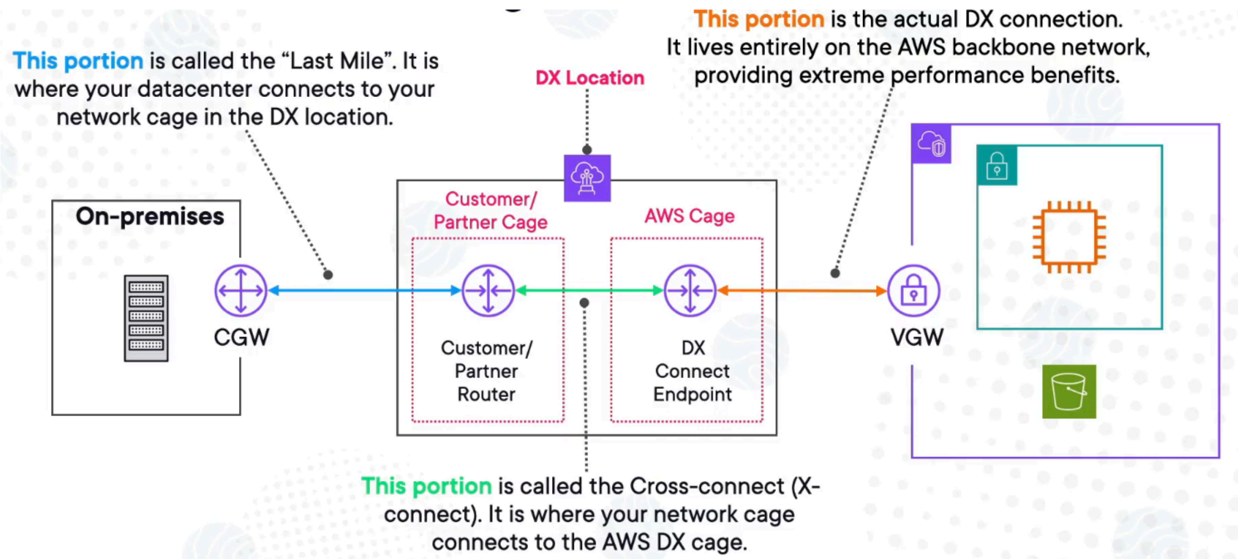
Advanced VPC: VPNs

- VPN: establishes encrypted connections between computer devices over public internet
    - Site to Site VPN
        - Must use publicly resolvable IP address on your side of VPN connection
        - If you have NAT-T (NAT Traversal) in place, then you use the public IP of the NAT device that fronts your CGW
        - You must enable route propagation within VPC route tables
        - Most secure due to IPSec connections
        - VGW (Virtual Gateway): VPN concentrator that is deployed on AWS side of VPN connection
            - You attach it to the VPC that you wish to create your site to site VPN connection with
            - Can customize Autonomous System Number (ASN)
        - CGW (Customer Gateway): Software appliance or physical networking device on the customer side of VPN connection
            - Required to fully configure device to work with site to site VPN, not all devices are supported
            - AWS CGW provides required information to AWS
    - AWS Client VPN: Managed client-based VPN service that enables secure access to AWS resources and resources in on-premises network
        - AWS-managed version of OpenVPN
        - Can use any OpenVPN compatible software to connect

        - Uses TLS and scales
        - Use this to access resources from any location
    - AWS VPN CloudHub: Allows you to securely communicate from one site to another via site to site
        - Allows you to set up and operate a hub-and-spoke model for multiple site to site (S2S) VPN
        - VPC is not required
        - Perfect for multiple data centers for low cost hub and spoke VPN connectivity
    - Third-party VPN
        - Typically obtained from AWS Marketplace and run on EC2 instances
        - Use cases
            - Need to enable transitive routing on AWS side
            - Need special capabilities not native to AWS managed VPNs
            - Bandwidth control
        - Need to disable source/destination check on EC2
        - Failover recovery is your responsibility
- Use Cases:
    - Securing hybrid cloud network architectures
    - Disaster Recovery networking connections
    - Establishing Transit VPCs
    - Interconnecting VPCs
    - Connecting to third-party vendors

Advanced VPC: Direct Connections, Direct Connect Gateways, and Transit Gateways

- AWS Direct Connect (DX): Establish a dedicated network connection from your premises to AWS
    - Can reduce network costs, increase bandwidth throughput, and provide more consistent experience
    - Have to be set up between datacenters and DX location
    - Must create a VGW and attach it to VPC
    - Offers private access to both public and private resources



- DX Connection Types
    - Dedicated: Physical Ethernet connection associated with single customer
        - Can be requested through DX console, CLI, or API
        - Best for maximum performance and full privacy but less flexibility for capacity
    - Hosted: Physical Ethernet connection that an AWS DX Partner provisions on behalf of customer
        - Request these by contacting a partner in the DX Partner Program
        - Lots of capacity flexibility, but need to choose the right partner for some capacities
    - Can take weeks to set up
- Virtual Interfaces
    - You leverage DX connections by attaching the DX Virtual Interfaces (VIF) to your VGW that is attached to VPC
    - VIF allow you to connect to AWS services over private DX connection instead of public internet
    - Public VIFs: S3, DynamoDB, Route 53, SQS
    - Private VIFs: EC2, RDS, Lambda, VPC endpoints (Services inside of VPCs)
- AWS Direct Connect Gateways: Globally available resource that allows you to connect multiple VPCs across different regions to your on-prem networks through a single Direct Connect connection
    - Direct Connect Gateways connect to a Direct Connect location in a region, which connects to the chosen DX location
    - VPCs must be in the same org
    - VPC to VPC does not work
    - Offers a centralized point for managing connections between on-prem network and AWS resources via AWS DX
- Traffic traversing Direct Connect connection is private, but not encrypted

- AWS Transit Gateways: Connects VPCs and on-premises networks through central hub, simplifies network, acts like a cloud router (Enables you to attach VPCs and VPN connections and route traffic between them)
    - Connect thousands of VPCs and on-prem networks
    - Control where traffic can go by using TGW route tables that are configured for attachments on VPCs
    - Integrates with DX connections, DX Gateway, and VPN connections
    - Can be shared with AWS RAM
    - Supports IP multicast, not supported by any other networking service
    - Can Peer TGW
    - Common TGW Attachments
        - VPC Attachment: Attaches VPC to TGW by picking one subnet from each AZ to be used by the transit gateway to route traffic
        - VPN Attachment: Allows you to connect on-prem networks via VPN to multiple VPCs over TGW
        - Peering Attachment: Used for both intra-region and cross-region TGWs and route traffic between them
    - TGW Route Tables: Each attachment for TGW is associated with exactly one TGW route table
    - Use Cases:
        - Network Segmentation: Isolated routing domains within a single TGW
        - Hub and Spoke: Centralized routing for multiple VPCs and on-prem networks
        - Shared Services: Route traffic to a shared service VPC from multiple VPCs
        - Complex Routing Scenarios

Advanced VPC, Miscellaneous Features and Scenarios

- VPC Flow Logs: Enables you to capture IP traffic information about any traffic going to and coming from network interfaces
    - Can be set up for Entire VPC, VPC subnet, or specific ENI
    - No performance impact
    - Also supports ELBs, RDS, Redshift, NAT Gateways, TGW
    - Can be sent to
        - CloudWatch Logs
        - S3: Cheap, long term storage
        - Amazon Data Firehouse: Autoscaling delivery system for analytics
    - Example: 5-tuple capture, each record captures within an aggregation interval (capture window)
        - 2 12345678910 eni-1235a6bc789101112 15.222.10.30 10.0.0.24 21234 443 6 29 4249 1418530010 1418530070 REJECT OK
        - VersionFlowLog(Default is 2) AccountID ENIID SourceIPAddress DestinationIPAddress SourcePort DestinationPort Protocol(TCP=6)

NumberOfPackets NumberOfBytes StartTime(Unix Time) EndTime (Unix Time) Action(REJECT/ACCEPT) LogStatus(OK/NoData/SkipData)
            - version, account-id, interface-id, srcaddr, dstaddr, srcport, dstport, protocol, packets, bytes, start, end, action, status
    - Can use Athena to query logs, works on S3 buckets and CloudWatch Logs
    - Can't modify existing flow logs
- VPC Traffic Mirroring: Copy network traffic from ENI and send traffic to out-of-band security or monitoring appliance
    - Set up source, filter, target, and session
        - Destination: ENI, Gateway, Load Balancer, or Network Load Balancer
        - Source and target do not need to be in the same account
- Egress-only Internet Gateways
    - Only useable by resources with IPv6 Address
    - NAT Gateway only for IPv6 resources
    - Prevents inbound initiation, allows outbound
    - No charge but data transfer has charge
    - Requires updates to VPC route tables
    - IPv4 and IPv6 is handled with different route rules within route tables
    - You attach Egress-only internet gateway to VPC just like normal IGW
    - ::/64 is the syntax for all IPv6 traffic
- Common Ports:
    - HTTP: 80
    - HTTPS: 443
    - SSH: 22
    - FTP: 21
    - Telnet: 23
    - SMTP: 25, 587 (with TLS)
    - DNS: 53
    - RDP: 3389
    - MySQL: 3306
    - PostgreSQL: 5432
    - Microsoft SQL Server: 1433
    - Redis: 6379
    - LDAP: 389

Advanced IAM: AWS IAM Identity Center and AWS Directory Services

- AWS Directory Services: Provides several options to set up and run Microsoft AD with other AWS services
    - AWS Managed Microsoft AD: Entire AD suite
        - Only one of the three that supports trust (one way or two way trusts)
        - AWS Enterprise services require 2 way trusts, Enterprise services are like Chime, AWS IAM Identity Center, and Workspaces
        - One-Way Trust: Users in one domain need to access resources in the other

        - Two-Way Trust: Access both ways
    - AD Connector: Directory gateway where you redirect requests to on-prem Microsoft AD, no caching (good for compliance)
        - Small and Large sizes
        - Can spread loads across multiple connectors
    - Simple AD: Standalone managed directory
        - Small and Large sizes
        - Subsets of features provided by AWS Managed AD
        - Cannot be joined with on-prem Microsoft AD
        - No MFA, trusts, Powershell, and certain service accounts
- AWS IAM Identity Center: Provides SSO for logins
    - Can connect to existing identity providers like AD, Okta, OneLogin
    - Most have AWS Organizations set up
    - Can use ABAC (Attribute-based Access Control) to set up fine-grained access controls into accounts
    - Can create and manage users directly in IAM Identity Center
        - Can grant users access to applications hosted in AWS
        - Can grant users access to AWS accounts in multi-account orgs
    - Compatible Applications:
        - Cloud apps like Salesforce, Microsoft 365, Confluence
        - SAML-2.0 business applications
        - EC2 Windows instances

Advanced IAM: Complex IAM Policies and Conditions

- Permission Evaluation Order
    1. Explicitly Deny
    2. Service Control Policy
    3. Resource Based Policy
    4. Identity Based Policy
    5. Permissions Boundaries
        a. Feature where you use a managed policy to set maximum permissions that an identity-based policy can grant to IAM entity
        b. IAM entity would only have permissions allowed by both identity based policies and its permissions boundaries (overlap)
    6. Session Policies
- Condition Statements:
    - ExternalID: Unique identifier when you assume a role in another account
        - Used to prevent confused deputy problem
    - aws:MultiFactorAuthPresent: Checks for MFA
    - aws:SourceIp: Can filter by IP
    - aws:PrincipalOrgID

Storage, Databases, Machine Learning, and Big Data Analytics

Relational Database Service

- Relational Database Management System (RDBMS)
    - Stores related data points in the form of tables
        - Tables: Stores logically connected data
        - Rows: Each row is an item
        - Columns: Each column is a field (attribute)
- Online Transaction Processing (OLTP): Primary purpose is to process database transactions, like processing orders, updating inventory, and managing customer accounts
- Online Analytical Processing (OLAP): Primarily meant to analyze aggregated data and generate reports, perform data analytics, or identify trends based on data
- Most commercial RDBMS uses StructuredQuery Language (SQL) to access database
- Relational Database Service (RDS)
    - Supports popular database engines, like SQL, PostgreSQL, Oracle, MariaDB, MySQL, Aurora
    - Managed service, can setup automated backups, manually create own backups, use IAM, VPC SGs and NACLs, Encryption at rest and in transit
    - DB Instance Classes: Determines computation and memory capacity
        - General purpose: db.m*
        - Memory optimized: db.z*, db.x*, db.r*
        - Compute optimized: db.c*
        - Burstable Performance: db.t*
    - DB Instance Storage Types
        - Provisioned IOPS SSD: I/O intensive workloads
        - General Purpose SSD
        - Magnetic: Not recommended for anything but supported
- Amazon RDS Networking
    - DB instances are created within VPC
    - Best practice to isolate DB instances within their own subnets
    - DB subnet Group: VPC subnets specifically to deploy DB instances
    - You control network access to RDS instances via VPC Security Groups
    - Option to designate if the DB instance gets a public IP (Almost never do this)
    - Cannot access with SSH and RDP
- RDS Backup and Maintenance
    - Specify backup window and set a retention period (can set to 0 to disable)
    - Backups are stored in S3 but you don’t have access to underlying buckets
    - Automated backups: First snapshot of a DB instance has full data, snapshots afterwards are incremental and only captures data changes
    - Can enable automated cross-Region backup replication
        - Not supported for Multi-AZ DB clusters

    - Manual Backups: Can keep around as long as you want
    - Cannot restore from a DB snapshot to existing instance, you need to create a new instance
    - DB instances uses lazy loading
    - You pay for RDs storage even if RDS instances is stopped
- Multi-AZ Deployments: RDS creates an exact copy in another AZ (called standby replica)
    - AWS automatically handles data replication between primary and standby instances
    - Multi-AZ DB Instance Deployment: 1 Standby DB Instance
    - Multi-AZ DB Cluster Deployment: 2 Standby DB instances
        - Big difference is that the standby instances can be used as read instances
        - Failover takes 1-2 minutes for both
    - For failovers and high availability
- Read Replicas
    - Read only copy meant to reduce load
    - Can be set up to be same AZ, cross-AZ, or cross region
    - Automatic backups must be on
    - Each read replica gets its own DNS endpoint
    - For read performance
- RDS Authentication
    - Password authentication (Basic)
    - AWS IAM authentication: Authentication token based on IAM credentials
    - Kuerberos authentication: External authentication via Kerberos and Microsoft AD
    - Integrates with Secrets Manager
- RDS Encryption
    - TLS by default for encryption in transit
    - RDS offers encryption at rest (encrypts all logs, backups, and snapshot)
        - Uses KMS key, must be enabled at instance creation, cannot change key once applied
    - Master DB instance must be encrypted for read replicas to be encrypted
    - If you need to encrypt Oracle or SQL Server DB Instance, you can use Transparent Data Encryption (TDE)
- Cost Optimization for RDS
    - You pay for:
        - RDS instances that are running
        - DB storage regardless of DB instance activity
        - Cross-region replication data transfers
        - Enhanced Monitoring (if you turned it on)
    - On Demand: Pay by the hour for the DB instance hours that you use
    - Reserved instance: You reserve capacity for 1 or 3 year term for discounts
    - No RDS Spot Instances
- RDS Custom

    - Only for Oracle or Microsoft SQL Server
    - Allows for OS and database customizations (more detailed setting)
    - Can allow access to underlying instances via SSH or Session Manager
- RDS Proxy
    - Allows applications to pool and share database connections
    - Improves ability to scale and resilience to failure, reduce memory and CPU overheard for opening new database connections, handles unpredictable surges
    - Useful for optimizing connections from AWS Lambda functions

Amazon Aurora

- Overview
    - Fully managed relational database engine compatible with MySQL and PostgreSQL only
    - Leveraged and managed via RDS (Engine type within RDS)
    - Built using AWS proprietary (not open-source)
    - Better performance than traditional RDS, but costs more (about 20%) than RDS
    - Can scale to large amounts of vCPUs and Memory
    - Deploy Aurora in Aurora DB Clusters
        - DB Cluster: Consists of 1 or more DB instances and a cluster volume
        - Cluster Volume: Virtual database storage volume that spans multiple AZs
        - DB Cluster Types: Primary, Replica
    - Endpoints:
        - Cluster Endpoint: Primary DB instance for a DB cluster
        - Reader Endpoint: One available Aurora Replicas
        - Custom Endpoint: Set of DB instances that you choose
        - Instance Endpoint: Connects to specific DB instance
- Aurora Storage
    - Automatically scales for you
    - 10GB - 128TB ranges
    - 6 copies of your data is copied (3 AZs, 2 in each) with auto recovery
    - Self healing, and can handle lots of failure before affecting availability
- Aurora Replicas
    - Only 1 primary (writer) instance
    - Can have up to 15 Aurora Replicas
    - Replicas connect to the same storage volume as primary DB (different from RDS read replicas), and like read replicas in RDS only support read operations and server as automatic failover targets, supports cross-region replication
- High Availability and Scaling
    - Aurora Auto Scaling dynamically adjusts number of Aurora Replicas
    - You can use automatic failover to a new primary instance or promote replica to become new primary instance (promoting replica is faster)
    - You customize the order in which replicas are promoted (0 (highest) to 15 (lowest))

    - More than one Replica can share the same priority, RDS promotes the replica largest in size
    - If no replicas exist, primary instance is recreated in the same AZ
- Aurora Backups
    - Automated backups are always enabled (like RDS, are continuous and incremental), and are stored in S3 like RDS
    - Can take manual snapshots
    - Default backup retention is 1 day (1 day is free), up to 35 days, unlike RDS minimum is 1 day instead of 0 days
        - Take a snapshot if you need something longer
    - Backtracking: Feature for Aurora MySQL-compatible DBs where you can rewind a DB cluster to specific time without restoring backups
        - Target Backtrack Window
        - Actual Backtrack Window
        - Limit for the window is 72 hours and causes a brief DB instance disruption
    - Aurora Cloning allows for faster and efficient database cloning
        - Quickly set up test environments
- Aurora Serverless: On-demand, auto scaling configuration for MySQL-compatible and PostgreSQL-compatible editions of Amazon Aurora
    - Good for infrequently, intermittent, and unpredictable workloads
    - Scales based on actual demand
    - Serverless Capacity Unit (ACUs)
        - Each ACU provides 2 GiB of memory, more memory means more compute and networking power (0.5 ACU to 128 ACU)
- Aurora Global Database
    - Recommended Disaster Recovery (DR) solution that spans multiple AWS regions
    - One primary AWS region, up to 5 read only secondary AWS regions
        - Each region can hold up to 16 read replicas
    - Replication data to secondary AWS regions are quick, and disaster recovery and promotion of regions is < 1 minute
- Aurora Machine Learning
    - Allows you to incorporate predictions via different ML servers with no experience required
    - Works with: Amazon Bedrock, Amazon Comprehend (NLP service), Amazon SageMaker (Managed ML service)
    - Use cases: Fraud detection, customized marketing and ads, recommended products to customers

Amazon DynamoDB

- Overview
    - NoSQL serverless database service that need consistent, single-digit millisecond latency
    - NoSQL tend to be unstructured and flexible

    - Automatic scaling
    - DynamoDB Tables (Regional)
        - Data stored in tables which have unique Primary Key (partition key)
        - Each row is called an item, and tables can have unlimited rows
        - Each item has columns called attributes
        - Each item can be up to 400KB in size
    - Schema Data Types
        - Scalar: Strings, numbers, binary, boolean, null
        - Document:Lists, maps (JSON documents)
        - Set: String set, number set, binary set
- DynamoDB High Availability
    - Automatically replicas across 3 AZs
    - Zero downtime maintenance
    - Automatically monitors tables and reports metrics to CloudWatch
    - CloudTrail can track API calls and related events
- DynamoDB Capacity Modes
    - Provisioned
        - You define amount of capacity required
        - Pay for provisioned capacity units (even if you don't use)
    - On-Demand
        - Flexible capacity
        - Pay per request
        - Good for spiky or unpredictable workloads
    - Strongly consistent: 1 read capacity units (immediate)
    - Eventually consistent: 0.5 read capacity units
    - If an item is over 4 KB, then round up to nearest capacity unit requirement
        - EX: 1 strongly consistent read for 5 KB item requires 2 RCUs
        - Read examples: GetItem, BatchGetItem, Scan, Query
        - Write examples: PutITem, UpdateItem, DeleteItem, BatchWriteItem
    - Scan: Reads every item in a table and returns all data attributes for every item
    - Query: Finds items based on primary key values
    - DynamoDB Auto Scaling: Leverages AWS Application Auto Scaling to dynamically adjust provisioned throughput capacity
        - Scaling policy specifies if you want to scale read or write or both
- DynamoDB Security
    - Publicly accessible service by default (public endpoint)
    - Encryption at rest by default via AWS KMS Keys
    - Can leverage VPC endpoints to keep traffic within AWS
    - Can leverage resource based policies for tables and indexes
- DynamoDB Global Tables
    - Feature that deploys multi-region, multi master database
    - Very low latency for replication (<1 second)
    - Made for Active-Active implementations
        - Means you can read and write to any of the tables within your regions

    - DynamoDB creates identical tables in your chosen Regions, and automatically propagates ongoing data changes to them
    - Must enable DynamoDB streams
- DynamoDB Streams
    - Captures time-ordered sequence of item-level modifications (creates, updates, or deletes)
    - Stored for up to 24 hours, near real time
    - StreamViewType: Configuration option (4 types)
        - KEYS_ONLY: Only key attributes of modified item
        - NEW_IMAGE: Entire item after modification
        - OLD_IMAGE: Entire item before modification
        - NEW_AND_OLD_IMAGES
- DynamoDB Accelerator (DAX)
    - In memory cache for DynamoDB
    - Helps speed up read requests with caching (up to 10x performance)
- DynamoDB Items Time to Live (TTL)
    - Cost-savings approach for deleting items
    - No write capacity units are used
    - Items are deleted within few days of TTL, not immediately
- DynamoDB Backups
    - DynamoDB
        - On-Demand Backups and point-in-time recovery (PITR)
            - PITR gives granularity and up to 35 days of recovery points
            - Charged based on size of table and duration of enablement
        - No impact on performance and availability
        - Backups are automatically encrypted
        - Charged based on size and duration of backups
    - AWS Backup: Can be used to implement automatic backup of tables
    - Imports and Exports
        - Imports creates a new table
        - Exports requires PITR and is useful for auditing

Other AWS Databases

- DocumentDB
    - MongoDB: Document database for scalability and flexibility (similar to NoSQL)
        - Data is stored via JSON documents
    - DocumentDB: Fully managed database service for MongoDB workloads
        - Uses same code, drivers, and tools that you use with MongoDB
        - Storage volumes automatically grows (up to 128 TiB in size, in 10 GB increments)
        - Infrastructure is extremely similar to Amazon Aurora
        - Can choose instance-based and elastic clusters
        - Supports up to 15 replica instances
        - Must be deployed in a VPC

        - Supports encryption via KMS keys
        - Offers point in time recovery for clusters
- Graph Databases
    - Stores nodes and relationships instead of tables and documents
        - Used for social networks, recommendation engines, knowledge bases/graphs (like Wikipedia), Fraud detection platforms
- Amazon Neptune
    - Fully managed, fast, reliable graph database service that makes it easy to build and run applications
    - Optimized for storing billions of relationships, can query with milliseconds of latency
    - High availability with read replicas and replication across 3 AZs, has point in time recovery and continuous backup
    - Supports Apache TinkerPop Gremlin, Neo4j's openCypher, and SPARQL
    - Can use for security, fraud detection, building connections between identities
    - Amazon Neptune Streams:
        - Logs every change in graph in the exact order with no duplicates
        - Can be used to notify people for certain changes, or maintain current version of graph data in another data store (S3, OpenSearch, ElastiCache)
- Amazon Keyspaces (For Apache Cassandra)
    - Cassandra: Open-source distributed database that uses NoSQL
        - Primarily used for big data (Has been used by Netflix)
    - Keyspaces: Amazon's Cassandra offering, fully managed and scalable
        - Serverless (you pay for what you use)
        - Auto scaling
        - Cassandra Query Language (CQL): Language specific to Cassandra
- Amazon Quantum Ledger Database (QLDB)
    - Ledger Database: NoSQL database that is immutable, transparent, and has cryptographically verifiable transaction log that is owned by one authority (has one source of truth)
        - Updates add a new record to the database
        - Used for cryptocurrencies, shipping companies, pharmaceutical companies
    - Amazon QLDB: Fully managed ledger database
        - Centralized component for logging transactions
        - Don't confuse with Amazon Managed Blockchain
        - Use PartiQL queries
        - Uses
            - Store Financial Transactions: Complete and accurate record such as credit and debit transactions
            - Supply Chain Systems: Provides details of every batch manufactured, shipped, stored, and sold
            - Maintain Claims History: Claims tracking and data integrity

            - Centralize Digital Records: Complete centralized record of employee details
- Amazon Timestream
    - Time-Series Data: Data points over time
    - Timestream: Fully managed database for time-series and LiveAnalytics data
        - Can store trillions of time series data points per day and analyze using built-in analytics functions
        - Keeps recent data in memory and moves historical data to cost optimized storage tier
        - Common integrations: AWS IoT Core, Amazon MSK, Amazon Kinesis, Amazon QuickSight, Grafana/Prometheus, Amazon SageMaker

Other AWS Storage, Transfer, and Migration Services

- AWS Snow Family
    - Provides petabyte-scale data collection and processing solutions at the edge (outside of AWS)
    - Made for migrating large-scale data into and out of AWS, with built-in computing capabilities
    - AWS Snowball Edge (Larger)
        - Considered Jack of all trades device
        - Storage Optimized, Compute Optimized, Compute Optimized with GPU
    - AWS Snowcone (Smaller)
        - Small, portable, and secure device for edge computing and data transfer
        - You can load data to device offline, or migrate via AWS DataSync online
        - Snowcone and Snowcone SSD
    - Data is loaded on device and sent to AWS
- AWS Storage Gateway - File Gateways
    - Storage Gateway: Hybrid cloud storage service that helps merge on-prem resources with cloud
        - Used for hybrid architectures
    - Set up local S3 File Gateway to allow NFS/SMB connections to S3 buckets
    - Easily upload your data to S3 and is performant, you use with S3 Lifecycle Policies and Glacier
    - Basically, you transfer data through NFS/SMB to S3 file Gateway, which is then transferred to an S3 bucket
    - FSx File Gateway: Same thing but for Amazon FSx for Windows File Server and if you need Windows-native SMB capabilities
- AWS Storage Gateway - Volume Gateways
    - Uses iSCI devices (completely different from NFS/SMB)
    - Perfect for backups and migration efforts
    - Cached Volumes and Stored Volumes options
    - Stored as EBS Snapshots inside of S3
- AWS Storage Gateway - Tape Gateways
    - Replaces backuping up to physical tapes

    - You use a Virtual Tape Library (VTL) that uses virtual tape cartridge
    - iSCSI devices but specific to tape drives
- AWS Lake Formation
    - Allows you to govern, secure, and share data for analytics and ML
    - You can create a data lake in S3
        - Data lakes: Centralized repo to store and process large amounts of data
    - Uses AWS Glue for ETL actions
    - One stop shop for big data
    - Workflows/blueprints: Contains complex multi-job ETL activities that does loading and updating of data in data lakes
        - Can be on-demand or scheduled
    - Can use tag-based access control
    - Can leverage: Amazon Athena, Amazon Quicksight, Amazon Redshift Spectrum, Amazon EMR, Amazon Glue
- AWS Transfer Family
    - Easily move data in and out of S3 or EFS using SFTP, FTPS, FTP (VPC Only), and AS2
    - Can create and manage users entirely within the service
    - Can also use other IDPs like AWS Microsoft AD, Okta, Amazon Cognito, and custom providers
    - Excels at bringing legacy application to the cloud, data is stored in S3 and EFS
    - Good for file sharing
- AWS DataSync
    - Solution for migrating large amounts of data from external storage to AWS
    - Move data between NFS and SMB shares
    - Agent-based or Agentless (AWS to AWS)
    - Agents are VM appliances used to read and write during a transfer and perform tasks
    - Can use S3, EFS, and FSx for storage
    - Can verify data integrity after transfers
- AWS Backup
    - Centralize and automate data protection and backups
    - Configure backup policies
    - Supports cross-region and cross-account
    - Backup Plans: Policy defining when and how you want to backup resources
        - Can specify policies based on tags
    - Backup Vault: container that stores and organizes your encrypted backups
    - PITR (Point-In-Time-Recovery): Supported for some resources
    - Backup Schedule: How often you want to run backups
    - Backup Window
    - Backup Period/Lifecycle: Number of days you want to store backups
    - Transitions: Can transition to cold storage after some time
    - Backup Vault Locks

        - Optional feature for additional security and control, helps enforce Write-Once-Read-Many (WORM) model
            - Only one: Each vault can have 1 vault lock in place
            - Compliance mode: Vault configuration cannot be altered or deleted by anyone including root
            - Governance Mode: Vault locks can be removed by users with enough IAM permissions
- AWS Application and Server Migration Services
    - Rehosting, Replatforming, Repurchasing, Refactoring/Re-architecting. Retire, Retain
    - AWS Application Discovery Service
        - Meant to help you plan your migration to AWS
        - Can track migration status, discovered servers, group into applications, collect usage and config data
        - Integrates with AWS Migration Hub
        - Agentless: Deploy Application Discovery Service Agentless Collector (Agentless Collector) via OVA file
            - Agentless Collector collects tons of information
        - Agent: Deploy AWS Application Discovery Agent on each VM and physical servers
    - AWS Application Migration Service (AWS MGN)
        - Replicates your source servers to an AWS account
- AWS Database Migration Service (DMS)
    - Easily discover all database, convert database schemas, and fully manage migrating different data stores to the cloud
    - You start by creating an AWS DMS server (EC2) that runs replication software
    - Then create source and target connection
    - You can allow DMS to migrate databases while keeping source DB online
    - Homogenous: Same type (Microsoft SQL to Microsoft SQL)
    - Heterogeneous: Different type (Oracle to Amazon Aurora)
    - Full load: Migrates data from source to target
    - Full load + Change Data Capture (CDC): Full data load while capturing changes on the source
    - CDC Only: Meant to only replicate data changes, good for keeping in sync
- Database Schema Conversions
    - DMS Schema Conversion: USed to assess complexity of migration, and convert schemas and objects
        - Objects that can’t be converted are marked for manual conversion
        - Only web version
    - AWS Schema Conversion Tool (SCT): Downloadable tool used to convert existing database schema from one DB engine to another
    - Only used for heterogeneous database engine transfers (homogeneous engine migrations don’t need this)

Big Data and Analytics Server

- Amazon Redshift
  - Big Data 3 V's: Volume, Velocity, Variety
  - Redshift is a managed, petabyte-scale data warehouse, based on PostgreSQL but used in big data applications and OLAP workloads
  - Provisioned:Collection of compute nodes and a leader node, which you manage directly
  - Serverless: Automatically provisions and manages capacity for you
  - Reserved Instances are available
- Redshift High Availability, Snapshots, and Performance
  - Redshift now supports Multi-AZ deployments
  - Single-AZ is default deployment
  - Use snapshots (can be automated or manual and stored in S3, and can be configured to be copied across regions automatically)
  - Load data into Redshift using large inserts for performance
  - Enhanced VPC Routing: Redshift forces all COPY and UNLOAD traffic between cluster and data through chosen VPC, allows use of VPC security access controls (Security Groups, NACLs, etc.)
- Amazon Redshift Spectrum
  - Use this to efficiently read/query from S3 using Redshift without having to load data first
  - Requires existing Redshift cluster already created
  - Spins up thousands of temporary nodes to process in parallel (massive parallel processing MPP)
- Amazon Elastic MapReduce (EMR)
  - Service to help with ETL processing
  - You create clusters that can contain hundreds of EC2 instances
  - Transform and load large amounts of data into and out of AWS
  - Hadoop Distributed File System (HDFS): Distributed, scalable file system for Hadoop that distributes stored data across instances
  - EMR File System (EMRFS): Extends Hadoop to add the ability to directly access data stored in S3 as it if was part of HDFS
  - Local File System: Locally connected disk created with each EC2 instance
  - Clusters: Groups of EC2 instances within EMR, each instance is a node
    - Primary: Manages cluster, coordinates distribution of data and tasks
    - Core: Run tasks and stores data in HDFS (typically long-running)
    - Task: Optional nodes that only run tasks
  - On-Demand: Most reliable and expensive option
  - Reserved: Minimum 1 year, cost saving, typically primary and core nodes
  - Spot: Cheapest, can be terminated with little warning, typically task nodes
- AWS Glue
  - Serverless data integration service for ETL workloads
  - Data Catalog: Metadata store with your table definitions, job definitions, and other information

    - Crawlers: Connect to data sources, infer data schemas, and create metadata table definitions in catalog
    - ETL Jobs: Extract data from sources, transform using Apache Spark scripts, and load into targets
    - Triggers: Resources or methods to start jobs
    - Glue Studio: Simplified visual interface to create and run ETL Jobs
    - AWS Glue Streaming: Allows you to handle streaming data in near real time
    - Job Bookmarks: Persisted state information of data that has already been processed by previous ETL jobs to prevent reprocessing of data
    - Glue DataBrew: Visual data preparation tool for cleaning and normalizing data
    - AWS Glue Elastic Views: Makes it easy to build views that combine and replicate data across multiple data stores
        - SQL: Can quickly create virtual tables in different sources
        - Automated: Copies data from source to target
        - Continuous: Looks for changes in source
- Amazon Athena
    - Serverless, interactive query service that makes it easy to analyze data in S3 using SQL, good for one-time queries
    - Supports CSV, JSON, ORC, Avro, Parquet
        - ORC and Parquet are recommended for optimal performance
        - Can be converted via AWS Glue
    - Athena Federated Query: Used when data is stored in other sources that aren't S3
        - Can query the data in place or build pipelines to extract data and store them in S3
        - Uses data source connectors (Lambda functions)
        - CloudWatch Logs, DynamoDB, DocumentDB, RDS, JDBC-compliant relational data sources, Athena Query Federation SDK for custom connectors
- Amazon Quicksight
    - Managed, serverless business intelligence (BI) data visualization service powered by ML
    - Integrates with RDS, Aurora, Athena, S3, and more
    - SPICE: Super-fast, parallel, in-memory calculation engine
        - Requires you to load data into Quicksight
    - CLS: If you have Enterprise offering, you can leverage Column Level Security (CLS)
        - Pay-per-session pricing for users that you place in a reader security role
    - You create separate users, Enterprise version allows creating groups of users, users only exist in Quicksight
    - Can share dashboards
    - Enterprise edition leverages ML for data insights, trends, and forecast metrics
- Amazon OpenSearch Service

    - Managed service that allows you to run search and analytics engine clusters, successor to ElasticSearch
    - Ability to search all fields within data
    - Provides dashboards for visualization of data
    - Security: AWS IAM, Amazon Cognito, KMS Encryption, TLS, AWS PrivateLink
    - OpenSearch Domain: AWS version of OpenSearch cluster
    - Access Policies: Resource-based, Identity-based, IP-based
    - OpenSearch Serverless: Auto-scaling option meant for infrequent, intermittent, or unpredictable workloads

Machine Learning Services

- Amazon SageMaker
    - ML service to quickly build, clean, train, and deploy ML models
    - SageMaker Studio: Web-based experience for running all ML workflows
    - Containers: Docker containers are used to build and runtime tasks
    - SageMaker Domain: Used to organize resources
    - Endpoint: Where you actually deploy a model
    - SageMaker Automated ML
        - SageMaker Canvas: Use ML to generate predictions without any code
        - SageMaker Jumpstart: Offers pretrained, open-source models
- Amazon Rekognition
    - Automates recognition of pictures and videos using DL and Neural Networks
    - Computer Vision Capabilities
    - Confidence Scores
- Amazon Polly
    - Text fo speech
    - Voice Options: Generative, Neural, Long-form, Standard Text-to-Speech (TTS)
    - Can use plain text of Speech Synthesis Markup Language (SSML) which gives you more control with tags
    - Pronunciation lexicons allows customized pronunciation of words
- Amazon Translate
    - Advanced ML service for automated language translation
    - Can be used via console or API, or uploaded in a document
- Amazon Lex
    - Service for building conversational interfaces for applications using voice and text
    - Easily build chatbots
    - NLU: Used to comprehend and interpret human language to know intent of text or speech
    - ASR: Used to convert spoken languages into text
- Amazon Connect
    - Allows you to quickly and easily scale your customer experience
    - AI-powered contact center
    - Commonly used with Amazon Lex for customer service flows
- Amazon Comprehend

    - Managed, serverless service that uses NLP to help understand meaning and sentiment in text
    - Amazon Comprehend Medical: Detects and returns useful information in unstructured clinical text
        - Run async operations on documents in S3 via StartPHIDetectionJob API, or synchronous operations via DetectPHI API
- Amazon Forecast
    - Managed service that uses ML algorithms to give very accurate time-series forecast
    - No longer available to new customers since 2024, AWS recommends using SageMake Canvas
- Amazon Kendra
    - Service that allows you to create an intelligence search service powered by ML and NLP
    - Can bridge between different silos of information ,allowing enterprises to have all required data in one place, with semantic and contextual understanding capabilities
- Amazon Textract
    - Managed service that uses ML to extract typed text, handwritten text, and data from scanned documents
- Amazon Personalize
    - Managed ML service that uses your own data to generate real-time recommendations for users
    - Good for personalized emails, targeted marketing campaigns, and personalized search results
    - Recommendations are primarily made based on interaction data
- Amazon Transcribe
    - Converts audio to text (Speech to text)
    - Supports batch transcription stored in S3 or real time streaming transcriptions
    - Can be used to identify the languages being spoken
    - Custom vocabulary filters allow modifying words in transcription output
    - Can use toxic speech detection
    - Redaction capabilities
- Amazon Fraud Detector
    - Managed fraud detection service, highly customizable based on your data

Scaling and Decoupling Architectures

Elastic Load Balancing

- AWS Global Infrastructure Review
    - Regions (Commercial and GovCloud)
    - Edge Location
    - Availability Zones

        - Reminder: AZ names in AWS accounts map to different AZ IDs in other accounts, AZ IDs map to same AZs
    - High Availability (HA) and Fault Tolerance (FT)
- Elastic Load Balancing Introduction
    - Automatically distributes incoming application traffic across multiple targets, like multiple instances across multiple AZs
    - Can setup custom DNS to point to an ELB for applications (Route 53 alias)
    - Monitor health of backend targets and only route to healthy ones
    - SSL/TLS termination for websites and applications
    - 4 Types: ALB, NLB, GWLB, CLB (Classic Load Balancer) : Avoid using
    - Schemes: Internet-facing (Public), Internal (Private)
    - Routing Algorithms
        - Round Robin (Default): Routes requests evenly in sequential order
        - Least outstanding requests: Routes request to targets with lowest number of in progress requests
        - Weighted random: Routes request evenly in random order, can use Automatic Target Weights (ATW)
- Application Load Balancers (ALB)
    - Layer 7 (Application Layer) load balancing
    - Only HTTP or HTTPS
    - Perfect for containerized applications
    - Listener: Checks connection requests from clients
    - Rules: When conditions for a rule are met, actions are performed
    - Target Groups: Each target group routes request to 1 or more registered targets
    - Rule Conditions: host-header, http-header, path-pattern, query-string, http-request-method, source-ip
    - Rule Actions: forward, redirect, fixed-response, authenticate-oidc, authenticate-cognito
    - Target types: EC2 instances, ECS tasks, Lambda functions, private IPs
    - Must have at least 1 SSL/TLS certificate for an HTTPS listener
    - Headers to know: X-Forwarded-For, X-Forwarded-Port, X-Forwarded-Proto
    - Port Mapping: ECS feature to dynamically assign target container ports
- Network Load Balancers (NLB)
    - Layer 4 (Transport layer) load balancing
    - TCP, TLS, UDP, TCP_UDP
    - Highest performing ELB
    - Allows public EIP assignment
- Gateway Load Balancers (GWLB)
    - Layer 3 (Network layer) load balancing, means IP packets
    - Meant for scaling and managing virtual network appliances
    - Specifically for the GENEVE protocol using port 6801
    - Examples: Intrusion Detection Systems, Intrusion Prevention Systems, Deep Packet Inspection
- Elastic Load Balancing Optimization

    - Session Affinity (Sticky Sessions): Feature that allows you to enable clients to always reach the same target
        - CLB, ALB, and even NLB
        - Uses a cookie to determine which target is used (Can control when it expires)
        - Perfect for maintaining user sessions (like maintaining shopping carts)
    - Deregistration Delay (Connection Draining)
        - Allows load balancers to keep existing connections open if instances are deregistered or become unhealthy
        - Can set delay timing from 1 second to 3600 seconds
    - SSL Certificates and HTTPS Listeners
        - Load balancers use X.509 certificates for TLS
        - Manage and integrate TLS certs using AWS Certificate Manager (ACM)
    - SSL Offloading
        - Offload: ELB listener is configured to use HTTPS/TLS, and connections are terminated at the ELB. Backend connections use HTTP
        - Pass-through: NLB listener is configured for TCP 443, so no encryption or decryption happens there. Connections are passed to backend resources
        - Bridging: ELB listener is configured to use HTTPS/TLS, and connections are terminated at ELB. Backend connections are established using a new SSL connection

Auto Scaling and High Availability

- Horizontal vs Vertical Scaling
    - Horizontal Scaling: Increasing number of resources/instances, common for web applications
        - Ex: One m5.large -> 3 m5.large
    - Vertical Scaling: Scaling resources to be larger, common for databases
        - Ex: m5.large -> m5.xlarge
- Auto Scaling Groups (ASGs)
    - Auto Scaling Group contains a collection of EC2 instances that are treated as a logical group for purposes of scaling and management, free to use
    - Offers self-healing and cost optimization for your groups of EC2 instances
    - Scale out: Add additional EC2 instances to handle increased workloads
    - Scale in: Remove unneeded EC2 instances
    - Easily set minimum, maximum, and desired amount of instances
    - Integration with ELBs
    - Launch Templates (LT): Specifies all required settings that go into building out an EC2 instance
        - Launch Configurations: Deprecated version with limited features and no versioning capabilities
- Auto Scaling Group Scaling Policies
    - Scaling Policies: Allow you to dynamically respond to changing user demands and workloads via horizontal scaling

    - Manual: simplest way to scale ASGs
    - Simple: Relies on different metrics for scaling, like CPU Utilization
    - Target Tacking: Scaling metric/value that should be maintained at all times
    - Step Scaling: Applies stepped adjustments to vary the scaling
    - Predictive: Forecast future capacity needs by historical data
    - Scheduled Scaling: Feature that allows you to set up automatic scaling for applications based on predictable load changes
    - Scaling Metric Examples
        - ASGAverageCPUUtilization
        - ASGAverageNetworkIn
        - ASGAverageNetworkOut
        - Custom
    - ASGs having a Scaling Cooldown after launching or terminating instances, pre-baked AMIs can have a lower cooldown
- Auto Scaling Group Health Checks
    - EC2: EC2 instance status
    - ELB: Healthy or Unhealthy status
    - VPC Lattice: Healthy or Unhealthy
    - EBS: Volumes are reachable
    - Custom
    - Grace Period: Configurable amount of time to delay health checks on instances, meant to allow apps to get up and running, execution of bootstrap scripts, etc.
- ASG Instance Maintenance Policies
    - You can create or alter an instance maintenance policy to affect EC2 Auto Scaling events, to optimize availability or cost
        - Launch before terminating: Better for availability
        - Terminate and launch: New instances are launched at the same time existing instances are terminated, better for cost
        - Custom Policy
- Auto Scaling Group Lifecycle Hooks
    - Lets you create solutions that are aware of events in the Auto Scaling instance lifecycle
    - Perform custom actions (or custom scripts) on instances when the corresponding lifecycle event occurs
    - Example Use Case: You need to make sure that instances created via an ASG check in with a 3rd party auditing tool when they launch or terminate
- Improving Service Connectivity using VPC Lattice
    - VPC Lattice: Managed application networking service to connect, secure, and monitor services for your application in a VPC
    - Meant to help reduce networking complexities

Containers and Images

- Containers and Images

    - Container: Standard unit of software that packages up code and all its dependencies so application runs quickly and reliably, from one computing environment to another
    - Dockerfile: Text document that contains all the commands or instructions that will be used to build an image
    - Image: Immutable file that contains the code, libraries, dependencies, and configuration files needed
    - Registry: Stores Docker images for distribution (private or public)
    - Container: A running copy of the image
    - Good for microservices, and lift-and-shift (package applications and move to the cloud as-is)
- Amazon Elastic Container Registry (ECR)
    - AWS service allowing you to store and manage Docker images
    - Has lifecycle management, cross-region and cross-account replication
    - Image Scanning: Identify software vulnerabilities (has Scan on push)
    - Encryption: Server-side encryption at rest using AWS KMS keys
    - Tagging and Versioning: Can make tags immutable to prevent overwrite
- Amazon ECS Overview
    - AWS service that allows you to easily launch and manage Docker containers running on AWS compute
    - ECS Cluster: Logical grouping of tasks (containers) or services that run on the capacity infrastructure that you set up for the ECS clusters
    - ECS Launch Types: EC2, Fargate, ECS Anywhere
        - EC2 Launch Type: Each EC2 instance must run the ECS Agent, AWS handles starting and stopping of containers, but you maintain EC2 instances
            - Best for price optimization, long running
            - Can use spot instances
        - Fargate Launch Type: Serverless, pay as you go where you don’t have to provision or manage any infrastructure
            - Create Task Definition, then ECS runs Tasks (containers) based on configurations
- Amazon ECS Tasks, Task Definitions, and IAM Roles
    - Task Definition: JSON document that contains blueprint for containerized application
    - Task Role: Each task can optionally have an IAM role attached, containers assume the role
    - Task Execution Role: IAM role that allows ECS container and Fargate agents to make API calls, used for pulling images from ECR or Secrets Manager
- Load Balancing Amazon ECS
    - Allows you to maintain a specified number of instances of a task definition within an ECS cluster, for high availability
    - ELB, ALB, NLB, and CLB are supported, avoid CLBs

    - Application Auto Scaling: SErvice for automatically scaling resources besides EC2
    - Auto Scaling uses metrics from CloudWatch to determine scaling, which ECS publishes
- Amazon ECS Storage
    - EBS: Databases, root volumes, ETL workloads
    - EFS: Data analytics, media processing, CMS applications (serverless)
    - FSx for Windows File Server: .NEt apps needing Windows file storage
    - Docker volumes: Sharing volumes on different containers on same host
    - Bind mounts: Mounting a host volume for data persistence
- Amazon Elastic Kubernetes Service (EKS)
    - Kubernetes (K8s)
        - Cluster: Consists of a control plane and set of worker machines that host containerized application
        - Node: Virtual or physical machine that hosts running applications within containers (like EC2)
        - Pods: Smallest deployable units of computing that you create and manage in Kubernetes
        - Service: Feature for exposing an application that is running as one or more Pods in cluster
        - Jobs: One-off tasks that run until completion then stop
        - Ingress: Object in Kubernetes that sets up external access to service, usually done via HTTP and load balancing
        - Storage: Persistent Volumes (PV) are independent objects for storage in a cluster
    - EKS Node Types
        - Managed Node Groups: Allows AWS to provision and manage nodes (EC2 instances)
            - Support spot instances
        - Self-manged Nodes: Nodes (EC2 instances) that you deploy with a compatible AMI
            - Support spot instances
        - AWS Fargate: Serverless offering where you don't have to provision, configure, or scale
    - Scaling
        - Kubernetes Metrics Server: Allows you to use Metrics API
        - Horizontal Pod Autoscaler: Kubernetes object that updates workload resources automatically
        - Cluster Autoscaler: Focuses on making sure that your cluster has enough nodes to schedule your pods, focuses on nodes not pods
- Amazon EKS Data Storage
    - EBS, EFS, FSx for Lustre, FSx for NetApp ONTAP, FSx for OpenZFS, S3
- Securing Amazon EKS
    - Service Account: Gives ability to assume an identity (IAM role)

    - EKS Pod Identity: How you assign an IAM role to your service accounts so they can make API calls to other services (Like EC2 instance profile but for Pods)
- Amazon EKS Distro (EKS-D)
    - Self-managed Kubernetes deployment
- Amazon ECS and Amazon EKS Anywhere\
    - EKS Anywhere: Managed by customer where EKS is managed by AWS
    - ECS Anywhere: Allows management of container-based apps on-prem
    - Amazon EKS Connector: Allows you to connect Kubernetes cluster to AWS
- AWS App Runner
    - Fully managed AWS service for deploying from source code or container images directly to a scalable and secure web app
    - App Runner handles everything else
- AWS App2Container (A2C)
    - CLI for lift-and-shifting applications into containers into ECS, EKS, or App Runner
    - Supports Java, ASP..NET

AWS Lambda

- Overview
    - Serverless, ease of use, event-based, pay as you go billing
    - Run code as virtual functions without having to provision or manage underlying servers
    - Runs up to 900 seconds (15 minutes), making them best for short running workloads
    - Free tier, integrates with AWS services, logging and monitoring with CloudWatch, runs in AWS owned VPC by default
    - Deploying Functions
        - Console: Directly upload all required code and dependencies
        - .zip: Upload an archive that contains function code and dependencies
        - Container Image: You build an image locally, upload it to ECR, and then Lambda pulls it
- Configuring Lambda Functions
    - Runtime: Language-specific environment your code will run in
        - Python, Java, .Net, Golang (Go), Ruby, Node.js, Custom Runtime
    - IAM execution role: If your Lambda needs an AWS API call, you'll use this
    - Networking: Optionally define a VPC, subnet, and SG for functions
    - Function Resources: Define amount of available memory allocated
    - Environment variables can be used
    - Leverage a /tmp directory
    - Integration to mount EFS available
    - vCPU scales automatically based on memory
    - Encryption helpers using AWS KMS keys to encrypt values
- Lambda Function Networking
    - By default, Lambda functions are deployed to an AWS-owned VPC that allows Internet Access

- Lambda VPC Use Cases
    - Access to private RDS/Aurora instances in VPC
    - Access to EC2-hosted applications, or applications on prem
    - Security requirement to restrict all internet access for services
- Lambda Function Concurrency
  - Concurrency: Number of in-flight requests that your function is serving at any time
  - Provisioned Concurrency: Number of pre-initialized execution environments allocated to your function, useful or reducing cold starts
  - Reserved Concurrency: Maximum number of concurrent instances allocated to your function, useful for ensuring critical functions can always run
  - Throttling: Functions will drop requests and fail if you run out of available concurrency (can request quota increase via AWS support)
- Lambda SnapStart
  - Provides as low as sub-second startup performance with no code changes
  - Available for: Java, Python, and .NET
  - No extra cost when using it for Java functions
  - Works by caching snapshots of the memory and disk state of function, skips the init phase of the environment lifecycle
  - Use cases: Latency-sensitive APIs, Latency-sensitive data processing tasks
- Important Lambda Features
  - Versions: Feature to publish copies of functions and manage deployment of the functions
  - What's included: Each version's code, runtime, architecture, memory, layers, and most other config settings
  - $LATEST: Current, unpublished version of your function, used to publish a new version
  - You can create aliases for functions
    - Pointers to a function version you can change and update
    - Access the targeted version using an alias ARN
    - Allows you to implement traffic splitting for canary testing
  - Lambda Layers: .zip file archive that contains supplementary code or data, typically use these to share common code or dependencies across Lambda functions

Event-Driven Architecture

- Amazon RDS Events
  - Near real time notifications containing information about events that occurred on monitored RDS instances
  - Can be configured to be sent to an SNS Topic, or EventBridge
- Amazon S3 Events
  - New object created events, object removal events, reduced redundancy storage object lost events, replication events
  - Supported Destinations: SNS, SQS, Lambda, EventBridge

    - Must grant S3 permissions to interact with destination
- Amazon EventBridge
    - Formerly CloudWatch Events
    - Serverless Event bus that allows you to pass events from a source to an endpoint
    - Events: Recorded change in an AWS environment, SaaS partner, or your own configured application/services
    - Rules: Criteria used to match incoming events and send them to appropriate targets
        - Event pattern: Define event source and pattern
        - Scheduled: recurring schedule
            - Rate-based rate, Cron-based
    - Event Buses: Router that receives events and delivers them to targets (Destinations)
- Amazon EventBridge Event Buses
    - Default Event Bus: Included within every AWS account, automatically receives events from AWS services
    - Custom Event Bus: Creates new, separate buses for customer workloads
    - Partner Event Bus: Meant to receive events from SaaS partner applications and services
- Amazon EventBridge Events
    - Event: Change in an environment (AWS account, SaaS partner service, app)
    - Some AWS services are “best effort” and some are “durable”
    - Rule: Specifies which events to send to which targets for processing
    - Archives: Allows you to create an archive of events to easily replay at a later time
    - Replays: You can specify which archive to use, the start and end time for events to replay, event bus to use, or one or more rules to replay events to
    - Sandbox: Allows you to experiment with creating event pattern matching, rule configs, and transformations of inputs
    - Schema: Defines structures of events that are sent to EventBridge
    - Schema Registry: Container for schemas, to collect and organize them into logical groups

Amazon API Gateway, AWS AppSync, and AWS X-Ray

- Amazon API Gateway Overview
    - Fully managed service that allows you to easily publish, create, maintain, monitor, and secure your APIs
    - Allows you to put a safe “front door” on your application
    - Integrates with Lambda, HTTP endpoints, and other services
    - Compatible with Swagger and OpenAPI
    - API Gateway Endpoint Types
        - Edge-Optimized: Default option, API requests sent through CloudFront edge location, best for global users
        - Regional: Same Region access, ability to leverage with CloudFront

        - Private: Only accessible via VPCs using interface VPC endpoints
    - API Gateway API Types
        - Rest API: Collection of HTTP resources and methods integrated with Lambda, HTTP endpoints, and other services
        - HTTP API: Integrates with HTTP endpoint and Lambda, simpler than Rest API, cheaper, minimal features
        - WebSocket API: Integrated with LAmbda, HTTP endpoints, and other services, but are invoked via frontend websockets
- Amazon API Gateway Authentication and Security
    - Resource policies control which specified principal to call the API
    - Can Deny API traffic based on source IP addresses or ranges
    - Authentication Options
        - Publicly accessible
        - IAM Roles/Users
        - Amazon Cognito User Pools
        - Custom Authorizer (AWS Lambda)
    - API Keys: Can provide API keys to customers for access
    - Usage Plans: Set who can access your deployed API stages and methods, and allows for throttling number of requests
    - To use a custom domain name, you must leverage ACM TLS certs for HTTPS
    - Edge-optimized endpoints require ACM certs to be created in us-east-1
    - Regional endpoints require ACM certs to be created in the same Region
    - Set up DNS using a Route 53 CNAMe or Alias Record (A)
- Amazon API Gateway Integrations
    - Example Scenario: You're creating a dynamic website with some static content, and want to minimize server maintenance and patching. Website needs to be highly available, and it will need to scale storage read and write capacity as quickly as possible to meet scaling demand
        - S3 to host static website
        - Amazon CloudFront can be set up to front the website hosted in S3, and security via AWS Shield
        - API Gateway connected to CloudFront
        - DynamoDB table with On-demand capacity to automatically adjust read and write capacity based on traffic, connected to API Gateway
        - Lambda functions integrated with API gateway for backend logic/process
- Amazon API Gateway Deployments and Versions
    - Create a Deployment to publicly leverage your new Rest API
    - Deployments get associated with a Stage
    - Stage: Snapshot of the API methods, responses, integrations, etc.
    - Enable caching, customize throttles, enable logging, enable canary testing
- AWS AppSync
    - Robust scalable, and serverless GraphQL interface for developers
    - GraphQL: Query language for APIs that enables apps to fetch data quickly
    - Combines data from multiple sources

- Debug Applications using AWS X-Ray
    - AWS X-Ray: Service that collects data about requests within applications, and gives you tools to view, filter, and gain insights about the data to identify issues and optimizations options
    - Perfect for insights: Collections application data for viewing, filtering, and gaining insights about requests and responses
    - Each portion of a request and response provide a trace from your application, which provide the ability to gain further insights
    - Good for tracing, integrates with many services

Serverless Authentication with Amazon Cognito

- Amazon Cognito Overview
    - AWS Native identity provider that offers authentication, authorization, and user management
    - Integrates with web and mobile applications
    - Users have the ability to sign in directly with username and password, or through third parties (Amazon, Google, Apple, etc.)
- Amazon Cognito User Pools
    - Directory of users that provide sign-up/sing-in options for application users, and provides authenticated users a JSON web token (JWT)
    - Adaptive authentication: Block suspicious sign-ins or add another second factor authentication based on identifying an increased risk level
- Amazon Cognito Identity Pools
    - Allows you to give your users access to services using temporary credentials
    - Users authenticate via third-party IdPs or even Cognito user pools
    - Leverages IAM roles and policies to grant required user permissions
    - Authenticated vs Unauthenticated
        - Authenticated: Validated and receive temporary credentials based on an assigned IAM role
        - Unauthenticated: Guest access where they have not logged in and you need to assign a default IAM role

AWS CloudFormation and Infrastructure-as-Code Services

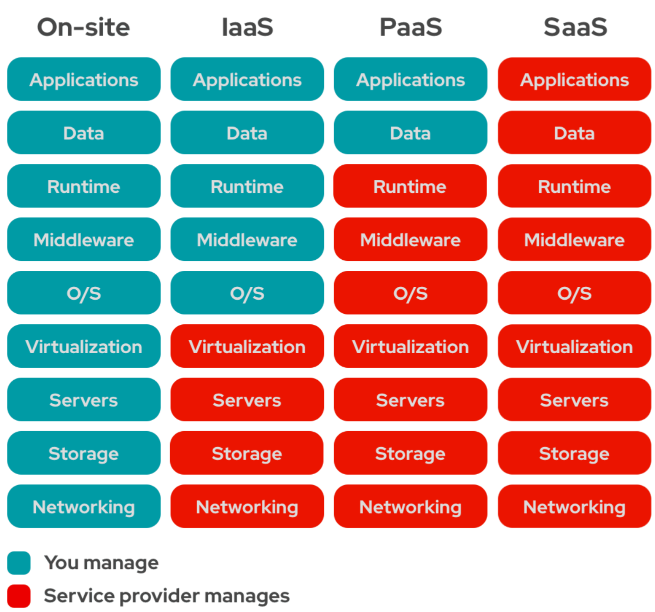
- Infrastructure as Code Review
    - Ability to provision and support computing infrastructure using code instead of manual processes and settings
    - Allows you to quickly duplicate environments across regions and accounts, avoid common errors when deploying resources, integrate with source control, implement a controlled and planned resource tagging approach
- CloudFormation Overview
    - AWS service that allows you to declare your AWS infrastructure as code for repeatable, automated deployments via “stacks”

    - Not every AWS resource is supported, but you can use custom resources for those that aren't
    - JSON or YAML
    - Required
        - Resources: Declare AWS resources you want to provision
    - Optional
        - Parameters: Used to input custom values to templates when you recreate or update a stack
        - Outputs: Output values from stack for easy reference
        - Mappings: Key-value pairs that are used to set values based on certain conditions
        - Conditions: Set conditions for which resources are created or configured
        - Transform: Macros that CloudFormat uses to process your template
    - Stacks are Regional resources
- AWS CloudFormation Service Roles
    - Service Roles allow the service to make API calls to resources for you
    - If no service role is set, temporary credentials from stack deployer are used
    - Use for least privilege principle
- Deleting AWS CloudFormation Stacks
    - Termination Policies: Feature that you can enable for each stack, used to prevent your stacks from being deleted
    - Stack Policies: Prevent specific resources from being updated/deleted
        - Allow updates using an Allow statement for a resource
        - One stack policy per stack
- Organizational Deployments via StackSets
    - Feature that enables you to create, update, and delete stacks across multiple accounts and Regions using a single operation
    - Managing Permissions
        - Self-managed: must create IAM roles that are used by the administrator account CloudFormation service to deploy your StackSets to different accounts and regions
        - Service-managed: Uses AWS Organizations to create the required roles used to deploy the stack instances across accounts and Regions, allows for automatic deployments for new accounts that join the organization
- Infrastructure Changes Using Change Sets
    - Feature that enables you to preview how all proposed changes to an existing stack might impact your current resources
        - Think `terraform plan`
- CloudFormation Custom Resources
    - CloudFormation method to write custom logic into CloudFormation templates
    - Used whenever you have logic that is complex and cannot be deployed or completed via built-in resource types
- CloudFormation cnf-init
    - Helper script in CloudFormation that helps define initialization tasks

- Usually defined in the “Metadata”: section of an EC2 resource
    - Use Cases: Getting metadata from CloudFormation, installing packages on EC2 instances, writing files to EBS volume at launch, starting/stopping and enabling/disabling system services
- Sharing Templates using AWS Service Catalog
    - AWS Service Catalog: Allows an organization to create and manage catalogs of IT services that they have deemed approved for using within AWS
    - Enables a company to centrally managed commonly deployed IT services
    - Good for IaC governance
- Self-Service Templates with AWS Proton
    - AWS Proton: Service that creates and manages infrastructure and deployment tooling for users
    - Automates IaC provisioning and deployments

AWS Elastic Beanstalk

- Reviewing Platform-as-a-Service (PaaS)
    - Complete development and deployment environment hosted in the cloud
    - You manage applications and develop, everything else is handled for you



- AWS Elastic Beanstalk Overview
    - Managed service meant to be a one stop shop for everything PaaS
    - You create a template of how you’d like your environments to appear, and it automates your deployments
    - Easily upload your code, test, and deploy multiple environments
    - Supports: Java, .NET, PHP, Node.js, Python, Ruby, Go, Docker, and tomcat
    - Service deploys multitiered, highly-available architectures, including EC2, ALBs, ASGs, and RDS
    - Good for migrating an application to AWS with minimal operational overhead

- AWS Elastic Beanstalk Environments
    - Application: Logical collection of Beanstalk components, including environments, versions, and configurations (similar to a folder)
    - Application Version: Points to an S3 object containing the code
    - Environment: Collection of AWS resources running your app version
    - Web Server Environment: Deploys a managed Auto Scaling group of web servers (EC2) behind an ELB with a CNAME URL
    - Worker Environment: Deploys a managed Auto Scaling group that scales based on messages in an SQS queue
- AWS Elastic Beanstalk Deployments
    - Single-instance: Deploys a single instance with an associated EIP for connecting to it (Great for development)
    - Load-balanced: Scalable environment using ELBs and ASGs for scalable and high-availability requirements (production)
    - All at once: Quickest method, can result in brief amount of downtime, application versions are deployed to all instances at one time
    - Rolling: Longer deployment time, no downtime, application versions are deployed one batch instance at a time
    - Immutable: Slow method, no downtime, application version updates are deployed to new instances that replaces old ones via a second ASG
    - Traffic splitting: For canary testing, great for testing health of new application versions while also still serving traffic to the old version
    - URL Swapping: Beanstalk allows you to do a Blue/Green deployment, by deploying the new version in a separate environment, then swapping the CNAMEs of the 2 environments

AWS Systems Manager

- Overview
    - Suite of tools designed to let you view, control, and automate nodes in AWS, other cloud providers, and on-premises
    - Nodes must be managed (SSM Agent installed and connected), with IAM permissions (AmazonSSMManagedInstanceCore AWS-managed policy)
    - SSM Agent
        - Required Amazon software that runs on your nodes and allows AWS SSM to update, manage, and configure the resources
- Patch Manager and Maintenance Windows
    - Patch Manager: Automates patching managed instances
    - Maintenance Windows: Definable schedule for performing actions on managed instances
    - Windows: Supports installing Service Packs on Windows nodes
    - Linux: Supports minor version upgrades on Linux nodes
    - Use patch policies and patch baselines to configure patching rules
    - Supports “Scan” and “Scan and install” patching operations
    - Patch Baselines and Patch Groups

        - AWS offers predefined patch baselines
        - Can create custom patch baselines
        - Custom allows for configuring categories and auto-approvals
        - Supports custom, alternative patch sources
- Automation and Documents
    - Tool that simplifies common maintenance, deployment, and remediation tasks
    - Create your own or use predefined automations
    - Automation: Tasks defined in a runbook that are performed by the Automation Service
    - Kicking off Automations: Schedule using Maintenance Windows, Trigger via EventBridge, Works with AWS Config rule remediations, Manually trigger via console or CLI
    - AWS Systems Manager Documents: AKA SSM documents, allows you to define actions that SSM performs on managed instances
        - Owned by Amazon
        - Owned by Me
        - Shared
- Run Command
    - Tool that allows you to remotely and securely manage managed nodes
    - Automate common administrative tasks and perform one-time configuration changes at scale, connects via SSM agent
    - Logging: Command logs can be saved to S3 and CloudWatch Logs
    - Trigger: Can manually trigger via console, SDK, or CLI, or EventBridge rules
    - Security: All API activity is captured via CloudTrail, and you restrict access via IAM
    - Important: Best for one-time tasks and requires that the SSM agent is configured and connected
- Parameter Store
    - Tool giving you secure, hierarchical storage for configuration data and secrets
    - Easily reference in scripts, SSM documents, and automations
    - Types
        - String: Any block of text you enter into the field
        - StringList: Comma-separated list of values, aka a list
        - SecureString: Any data stored that needs to be encrypted, direct integration with AWS KMS
    - Parameter Store vs Secrets Manager
        - Parameter Store: 4Kb Standard/ 8Kb Advanced, Broader use cases, No automatic rotations, Free for standard parameters
        - Secrets Manager: 64Kb, Specifically meant for secrets, Built-in automatic rotations, never free

Amazon SQS and Amazon SNS

- Decoupled Architectures

- Tightly Coupled Architecture: Components heavily depend on each other, if one is impacted/suffers downtime
- Decoupled Architecture (Loosely Coupled)
- Push (Topic): Messages are sent by a producer to a server (topic) and immediately sent (pushed) to a consumer
  - SNS
- Pull (Queue): Messages are sent by a producer to a server (queue), queued up, and then get pulled off by a consumer
  - SQS

- Amazon SQS Overview
  - Messaging queue service that allows asynchronous processing of work
  - One resource (producer) sends a message to an SQS queue, and then another resource (consumer) retrieves that message
  - Send messages to the queue via SendMessage API
  - Poll for and receive messages via the ReceiveMessage API
  - Use DeleteMessage API after receiving and processing a message
  - Access Control and Security
    - AWS IAM and Access Policies: Can leverage IAM for securing SQS queues
    - Encryption: Supports server-side encryption (SSE), SQS-managed encryption keys (SSE-SQS) or SSE-KMS
    - Public Endpoints: Public service by default, can restrict access via VPC endpoints
- Amazon SQS Queues
  - Standard and FIFO queue types
    - Standard: Nearly unlimited API calls/messages
      - "At least once" message delivery
      - Not guaranteed to be delivered in order
    - FIFO: Use if order of operations is critical, or where duplicates can't be tolerated
- SQS Queue Attributes
  - Delivery Delay/Message Timer: Delay the delivery of a new message for specified time
  - Message Retention: Amount of time that SQS retains message, default 4 days
  - Short-polling: Default option where "ReceiveMessage" calls do not wait to poll queues
  - Long-polling: Non-zero wait time for "ReceiveMessage", helping reduce number of API calls and empty responses
  - Visibility Timeout: Length of time a message can be received from a queue
    - Message will remain invisible to other consumers
    - If message is not deleted before timer, message reappears in the queue
  - Can use SQS Extended Library for larger sizes
- Dead Letter Queues
  - DLQ are targets for messages that cannot be processed successfully

    - Technically just other standard or FIFO queues
    - Useful for debugging, troubleshooting, or redrive which allows you to move messages back to source queue or into another queue
- Amazon SNS Overview
    - Push-based messaging is where messages are sent by a producer to a server and are immediately sent (pushed) to a consumer
    - Producers send messages to an SNS topic
    - Consumers subscribe to SNS topic to receive messages, messages encrypted by default
    - Can use SNS Extended Library for larger sizes
    - Can use IAM to control access, and access policies assigned to individual SNS topics
- Amazon SNS Topics
    - Standard vs FIFO Topics
        - FIFO offers strict messaging ordering with exactly-once delivery
        - FIFO topics can only send messages to SQS FIFO queues
    - SNS topics support cross region and cross account delivery
- Fanning out with SNS and SQS
    - SNS Fanout: Multiple subscribers on the same topic so messages are sent to multiple endpoints at the same time
    - Fanning out provides decoupling and scalability, by allowing multiple consumers to process messages independently

Amazon Kinesis

- Overview
    - Allows you to ingest, process, and analyze realtime streaming data and video
    - Considered a polling-based/pull-based approach
    - Perfect for real time or near real time processes
- Amazon Kinesis Data Streams
    - Useful for real time streaming of incoming data
    - Applications (consumers) read data records off the streams
    - Perfect for streaming applications that require large amounts of fast data ingestion
    - Producers: Constantly pushing data to Kinesis Data Streams
    - Consumers: Applications that processes the data in real time
    - Kinesis data streams contain shards, which have sequenced data records
        - Data records: Unit of data stored in streams (up to 1MB)
        - Shard: Streams are made up of 1 or more shards that have a set amount of capacity
    - Supports encryption using KMS
- Kinesis Data Stream Consumers and Producers
    - Producers: Amazon Kinesis Producer Library (KPL)
        - Meant to make ingesting data to a data stream simpler for applications
    - Consumers: Kinesis Client Library (KCL)

        - Standalone Java software library
        - Simplifies consuming and processing data from Kinesis Data Streams
    - Enhanced Fan-Out
        - Allows for push-based system designs, default for Kinesis Data streams is pull-based
        - Allows data to immediately get pushed to consumers once it is ready
        - Allows for extremely high throughput
        - Used when you need to scale consumers of a Kinesis Data Stream
- Data Stream Capacity Modes
    - Data stream capacity: Configuration to set how much data a stream manages and can handle
    - On-demand: Service automatically scales and manages your shards
        - Good for highly variable and unpredictable application traffic
    - Provisioned: You actually set the number of shards that you want to be created, can manually increase and decrease number of shards
- Amazon Data Firehose Overview
    - Near real time, fully managed data streaming solution
    - Producer sends data to Data Firehose, and automatically sends it to configured destination, like S3, Redshift, OpenSearch, and 3rd party providers
    - Has automatic scaling, serverless, can use PutRecord and PutRecordBatch API calls, allows for data transformation mid-flight via Lambda
    - Amazon Kinesis agent: Java app that sends data to Firehose streams
- Firehose Streams
    - Direct PUT: Your producer applications write directly to the stream using PUT commands
    - Amazon Kinesis Data Streams: Easily configure a data stream as a data source to read data and then load to chosen destination
    - Amazon MSK: Use MSK as a data source
    - Commonly used for sending several formats of documents
    - Can leverage Firehose for converting Parquet and ORC formats, and compressing records, and supports easy data transformations using Lambda
- Amazon Kinesis Data Analytics (Now Amazon Managed Service for Apache Flink)
    - Easily process and analyze streaming data using SQL
    - Applications read and process streaming data in real time
    - Also allows you to perform conversions, transformations, enrichments, and filtering using preprocessing Lambda functions
    - Kinesis Data Analytics for SQL Applications is deprecated and discontinued
- Amazon Kinesis Video Streams
    - Managed service which is meant to stream and store live video from devices to AWS
    - Useful for real time video processing or batch video analytics

Amazon MSK and Amazon MQ

- Amazon Managed Streaming for Apache Kafka (MSK)

    - Fully managed service for running data streaming applications that leverage Apache Kafka
    - Provides control-plane operations and you can create, update, and delete clusters as required
    - You deploy clusters in a VPC across multiple AZs
    - MSK will manage the cluster broker nodes and Zookeeper nodes
    - Offers an alternative to Amazon Kinesis
    - Recovery: Offers automatic detection and recovery from common failure scenarios for cluster nodes with minimal impact
    - Encryption: Supports TLS and KMS for encryption at rest
    - MSK Serverless:Cluster type within Amazon MSK offering serverless cluster management with automatic provisioning and scaling
    - Common consumers: Custom applications running on EC2, ECS, and EKS, and managed services like Kinesis Data Analytics and AWS Glue
- Amazon MQ
    - Message broker service allowing easier migration of existing applications to AWS
    - Supports Apache ActiveMQ and RabbitMQ
    - Commonly used to send ingested between large, cloud native applications
    - Amazon MQ for ActiveMQ: Active/standby deployments have one instance that will remain available at all times
    - Amazon MQ for RabbitMQ: Cluster deployments that are logical groupings of 3 broker nodes across multiple AZs sitting behind an NLB

Amazon Step Functions and AWS Batch

- AWS Step Functions
    - Serverless orchestration service for business applications and workflow
    - Primarily used to orchestrate Lambda functions, supports integrations with SQS, API Gateway, and others
    - Allows retries, parallel processing, can require human approval, built in error-handling
    - Use Cases: Data processing workloads, order processing, security automation with human approval
    - Different States
        - Pass: passes any input directly to its outputs/appends to a fixed data set
        - Task: Single unit of work performed
        - Choice: Branching logic
        - Wait: Time delay
        - Parallel
        - Map: Set of steps based on elements of an input array
        - Succeed and Fail
    - Standard (Long running) and Express (High event rate) workflow types
- AWS Batch
    - Fully managed batch processing service
    - Run batch computing workloads within AWS that run on EC2 or ECS/Fargate

- Automatically launches any required compute as you need it
- You submit jobs as needed or schedule them
- Jobs: Units of work that are submitted to AWS Batch (shell scripts, executables, docker images)
- Job Definitions: Specify how your jobs are to be run (blueprint for the resources in the job)
- Job Queues
- Compute Environment
    - Managed vs Unmanaged
    - Fargate (Recommended) vs EC2 (more control)

Application and Network Caching

- Amazon CloudFront Overview
    - Content delivery network (CDN) service that securely delivers data, videos, applications, and APIs
    - Reduces latency and provides higher transfer speed using AWS edge locations
    - You set up origins that host your content, and Cloudfront handles caching content
- Amazon CloudFront Origins
    - Origin: Location of any stored content (S3 or custom)
    - Distributions: Configuration telling CloudFront which origin server to use and how to manage delivery of cached content
    - S3: Easily front an S3 bucket, Grant access and customize security via an Origin Access Control (OAC)
    - Custom Origin: Supports LABs, EC2, any reachable HTTP server
    - CloudFront is only for HTTP/HTTP connections (Layer 7 traffic)
    - Time-to-live (TTL): Time files live in cache
    - Invalidation: Can manually invalidate files to remove them before expiration time and force an origin fetch
    - Versioning
- Amazon CloudFront Security
    - Origin Access Controls (OACs)
        - 2 ways to control access to S3 origins
            - Origin access identity (OAI)
            - Origin access control (OAC): Recommended
    - AWS WAF: Can create AWS WAF web ACLS and associate them to CloudFront distributions
    - AWS Shield: Mitigate and prevents DDoS attacks
    - Georestriction (Geoblock): Prevents users from accessing content from specific geographic locations (allowlist or blocklist)
- CloudFront Custom Domain Names and TLS
    - Viewer Protocol Policy: Configure protocol policy that you want viewers to use to access content (Ex: “HTTP and HTTPS”)
        - Can set to require HTTPS

    - Custom DNS TLS: ACM certs (must be deployed in us-east-1) can be used with CloudFront
- CloudFront Functions and Lambda@Edge Functions
    - CloudFront Functions
        - Lightweight functions managed within CloudFront
        - Only support JavaScript
        - Perfect for high-scale, latency-sensitive CDN customization requirements
        - Very fast but has less resource power
        - Only works at viewer request and response portion
    - Lambda@Edge Functions
        - Managed within LAmbda and support Node.js or Python
        - Must be deployed to us-east-1 region
        - Works on all 4 locations (Viewer response, viewer request, origin response, origin request)
            - Use cases: Performing basic auth and authorization closer to the users
            - Customize content based on user location, device, or preference
            - Add security headers or enforce HTTPS
            - Optimize content delivery by compressing or converting files
            - Basically: Any custom logic closer to end users
- AWS Global Accelerator
    - Networking service that sends traffic through AWS's global network infrastructure via accelerators that send traffic to endpoints
    - Meant to increase performance/decrease latency
    - Helps with IP caching and white listing by using 2 Anycast IPs that send traffic to edge locations and then to your applications
    - Differences from CloudFront:
        - Meant for TCP or UDP traffic
        - Great for static IP requirements
        - More control over failovers
    - Supported Endpoint Resources: Elastic IPs, Amazon EC2s, ALBs, NLBs
- AWS Global Accelerator Security and Health Checks
    - Standard accelerators automatically perform health checks (Traffic is only sent to healthy endpoints)
    - Uses Route 53 health checker data
    - ALB/NLB/EC2 enables internet traffic to reach them directly via endpoints
    - Must have an IGW in place for VPC
    - Don't need public IP Addresses on ELBs or EC2s
- Amazon ElastiCache
    - Managed distributed in-memory caching service for: Memcached, Redis, Valley
        - Memcached is for simple data types and simpler uses, other 2 are for more complex uses and has more features
    - Helps meet high performance, low-latency application requirements
    - Great for heavy and repeated read requests

- Use Cases:
    - Caching for Web Applications:Cache database query results, session data, or API responses
    - Session Stores: Manage user sessions in a distributed web applications, which allows you to maintain consistent session data across multiple servers
    - Online Gaming Leaderboards: Maintain an online leaderboard for a multiplayer game, updating player scores and rankings in real time

Miscellaneous Services

- Amazon Pinpoint
  - Enables you to engage with customers through variety of different messaging channels
  - You create journeys to customize engagements with consumers
  - Primarily intended for Marketing teams and Business users
- AWS Amplify
  - Offers tools and libraries for front-end-web and mobile developers to quickly build full-stack applications on AWS
  - Offers: Frontend libraries, UI Components, Backend building, Frontend hosting
  - Provides easy AuthN and authZ for applications
- AWS Device Farm
  - Application testing service for testing and interacting with Android, iOS, and web apps
  - Usable on actual phones and tables hosted by AWS
  - Automated and Remote access testing
    - Automated: Scripts or built in tests
    - Remote Access: Swipe, gesture, and interact in real time via web browsers
- AWS Wavelength
  - Build applications that require edge computing infrastructure
  - Also used to increase resiliency of existing edge applications
  - Typically used for specific 5G networking requirements
- Amazon AppFlow
  - Fully managed integration service for securely exchanging data between SaaS applications and AWS services
  - Example: Salesforce to S3
  - Specifically designed to integrate with SaaS sources for data ingestion, offloads operational overhead and improves application performance
- Amazon Simple Email Service (SES)
  - Allows you to use own email addresses and domains
  - Large-scale email solution

Securing and Monitoring Workloads

Amazon CloudWatch

- Overview
    - Monitoring and observability platform for insights
    - THE monitoring and logging service
    - System Metrics
    - Application Metrics
    - Logging
    - Alarms
    - Insights
- CloudWatch Logs
    - Can query logs
    - Encrypted by default, and can use custom KMS keys
    - Can send logs to S3, Kinesis Data Stream, Data Firehose, Lambda and AWS OpenSearch
    - Log Events: Record of what actually happened, containing timestamp and raw event message
    - Log Stream: Collection and sequence of leg events from same source
    - Log Group: Collection of log streams
    - Can set log retention period
    - Log Sources: SDK, CloudWatch Agent, Lambda, VPC Flow Logs, Gateway APIs, CloudTrail, Route 53 DNS, Elastic Beanstalk, ECS Containers
    - CloudWatch Logs Subscription Filters: Gives you real-time feed of log events that you can have delivered to other services
        - Can configure filter patterns to determine what logs get forwarded
    - Can export logs to S3 (can be different account)
        - Can take up to 12 hours to be available for export
    - Regional Resources
- CloudWatch Logs Agent
    - Default behavior is to not send logs to CloudWatch Logs
    - Installable agent that allows collection of system-level metrics and logs on EC2, on-prem servers, and custom metrics from StatsD and collectd
    - Versions
        - CloudWatch Agent: Original version with limited support
        - CloudWatch Unified Agent: Updated version with more options
    - RAM usage and Process information is supported through the Agent for EC2, basically Agent provides much more information than default
- CloudWatch Metrics
    - Regional
    - Most services provide free metrics, like CPU for EC2 (default is 5 minute interval)
        - Detailed monitoring is optionally available for some resources
            - Additional costs

            - For EC2, metrics are delivered from 5 minute intervals to 1 minute intervals
        - Can publish custom metrics
        - Can create and share dashboards containing view of metrics
    - Namespace: Container to group and isolate metrics
    - Time Stamp
    - Dimension: Name/value pair to identify metric (Like instance ID)
    - Metrics Streams: Continually stream CloudWatch metrics to destinations (similar to subscription filters but for metrics)
        - Commonly used to stream to S3
        - Common 3rd party is supported
- CloudWatch Alarms
    - Alarms watch a metric over a specific time period and change state based on thresholds
    - Allow you to automatically start actions on your behalf
    - Metric Alarm: Watches single metric, can perform 1 or more actions
    - Composite alarm: Monitor alarm states of multiple other alarms using AND/OR logic
    - States: INSUFFICIENT_DATA, OK, ALARM
        - You can set your alarm to trigger when it goes into an OK state if you want
    - CloudWatch Alarm Actions: Send notification to SNS, EC2 Action, EC2 Auto Scaling Option
- CloudWatch Insights
    - Container Insights: Collect, aggregate and summarize metrics and logs
        - EC2, EKS, Fargate, Kubernetes
    - Application Insights: Help detect problems with applications
        - Powered by SageMaker for automated dashboards
        - Helps troubleshoot issues across entire tech stacks
        - Can generate EventBridge Events
    - Lambda Insights
        - Summarizes system level metrics including CPU time, disk, and network
        - Uses CloudWatch Lambda extension
    - Contributor Insights
        - Analyze log data from applications and create time series for contributor data
        - Top-N contributors, unique contributors, usage
        - Contributors don't need to be users, can be URLS, hosts, etc. and can be used to identity bad hosts, heavy network users, URLs with the most errors

Miscellaneous Logging and Monitoring Services

- Amazon Managed Service for Prometheus

  - Serverless, Prometheus compatible service for securely monitoring container metrics, auto scaling, across 3 AZs
  - PromQL query language for exploring and extracting data
  - Stored in workspaces for 150 days
  - Commonly used to set up advanced monitoring of EKS or self-managed Kubernetes clusters
- Amazon Managed Grafana
  - Similar to Prometheus but for Grafana, AWS does auto scaling, setup, and maintenance
  - Used to secure data visualizations for instantly querying, correlating, and visualizing operational metrics, logs, and traces from different sources
  - Workspaces allow for separation of data visualizations and querying
  - Integrates with different sources including: CloudWatch, Amazon Managed Service for Prometheus, Amazon OpenSearch Service, and Amazon Timestream
  - Does "Single Pane of Glass" type dashboards
  - IoT devices monitoring is a common use case

AWS Organizations and Multi-account Architectures

- Multi-account Strategies
  - Structure controls for security and compliance
  - Isolated workloads
  - Billing separation
- AWS Organizations
  - Free governance tool that allows you to create and manage multiple AWS accounts
  - Main account is called the management account aka payer account
  - Other accounts that join the org are called member accounts
    - Member accounts can only belong to a single org
  - Logically group and manage member accounts using Organizational Units (OUs)
    - Root OU has everything
  - IAM Identity Center (SSO) commonly integrates with AWS Organizations
  - aws:PrincipalOrgID is a global condition key that is used in policies
- Features Sets
  - Consolidated Billing: Fee feature that allows all member account bills to roll up to single payer account
    - Can share reserved instance discounts and saving plans discounts
    - Can combine usage across all member accounts to share volume pricing discounts
  - All Features:
    - StackSets: Deploy CFN stacks across accounts and regions
    - Delegated admins: Delegate policy/service management to member accounts
    - OrganizationAccountAccessRole: IAM role created in each member account

        - Service Control Policies (SCPs): Policies to react/allow account actions
    - Can have centralized CloudTrail logging, can avoid running into service account limits (if you hit the limit of VPCs in an account, you can create another account to increase the limit), centralized CloudWatch logs
- Service Control Policies (SCPs):
    - JSON Policy that gets applied to accounts and OUs to restrict actions
    - Do not grant permissions, just dictate permissions that can be granted via IAM
    - You need to explicitly allow permissions, default behavior is to deny actions
- AWS Control Tower
    - Simplified, quick way to set up and govern AWS multi-account environment
    - Uses AWS Organizations to automate account creation, detect governance drift, implement security controls
    - Requires IAM Identity Center enabled
    - Landing Zone: Multi-account environment for all AWS resources, allowing enforcement of compliance regulations on all member accounts in the org
    - Shared accounts: AWS Control Tower automatically creates Log archive and Audit accounts (shared accounts) within a security OU that servers as a central logging account for CloudTrail and auditing configurations
    - Controls (Guardrails): Control Tower automatically applies mandatory preventive controls and effective controls for securing the org
        - Preventative: Implemented using SCPs, prevents actions
        - Detective: Implemented using AWS Config rules, not preventative
            - COMPLIANT or NON_COMPLIANT, if NON_COMPLIANT you can trigger remediations (Lambda or SSM) and notifications (EventBridge + SNS)
    - Control Tower Account Factory
        - Feature that enables cloud admins and users in IAM Identity Center to provision accounts in landing zone
- AWS Resource Access Manager (RAM)
    - Free service that allows sharing resources with other accounts
    - Can be member accounts in org or external accounts (by sending invites)
    - Commonly shared resources: Transit Gateways, VPC subnets and security groups, AWS Network Firewall policies, Route 53 resolver rules, VPC customer managed prefix lists

Account Security and Governance

- AWS CloudTrail (Regional)
    - Records user and resource activity, records AWS Management Console actions and API calls (enabled by default and can't be turned off)
    - Sources: Management Console, AWS CLI, AWS Service Actions, AWS SDK API Calls
    - CloudTrail Event: Record of an activity in an AWS account, providing historical trails

        - Management Events: Example is TerminateInstances EC2 API call (Default on)
        - Data Events: Resource operations performed on or in a resource
        - Insights Events: Unusual API call/errors
        - Stored for up to 90 days, can customize to store within CloudWatch and S3 if you need more time
- CloudTrail Trails (Regional)
    - Captures events in respective regions
    - Supports multi-region
    - Customize delivery of logs for long term storage in CloudWatch and S3
    - Trigger rules based on EventBridge events
    - AWS KMS encryption supported
    - Can integrate with SNS
    - Organizational Trail: Enables delivery of events in management account and all member accounts in an organization to same S3 bucket, CloudWatch Logs, and Amazon EventBridge
    - Log file validation: Feature that tells you if a log file was modified, deleted, or unchanged
- AWS Config
    - Tool that allows you to view configuration history of your infrastructure over a period of time
    - Can create rules to make sure resources conform to requirements, per region
    - Can receive alerts via SNS
    - Works well with AWS Organizations to aggregate results across regions and accounts
    - Integrates with S3 for storing config data
    - Statuses: COMPLIANT and NON_COMPLIANT
    - Reactive service, not proactive, only alerts you and records info
    - Can evaluate tagging compliance
    - Common use cases: restricted-ssh, ec2-ebs-encryption-by-default, s3-bucket-server-side-encryption-enabled, cloudtrail-security-trail-enabled
- AWS Config Rules and Remediations
    - AWS Config Rule: Compliance check helping you manage configuration settings for specific resources,which then evaluates compliance status
    - AWS Managed Rules or Custom Rules
        - Can be triggered based on config changes, or scheduled
        - Custom rules leverage AWS Lambda or Guard for evaluation
        - You can perform remediations with retries
            - Done via AWS System Manager Automation documents
    - Never free
- AWS Trusted Advisor
    - Auditing tool for AWS accounts
    - Requires no agents, installation, etc.

    - Makes recommendations to save money, improve availability, performance, security gaps
    - Check Categories: Cost optimization, Performance, Security, Fault tolerance, Service limits, Operational excellence
- Amazon Inspector
    - Automated security assessment service that helps improve security and compliance
    - Assess for vulnerabilities or deviations from best practices
    - Findings are sorted by level of severity
    - Findings can be sent directly to Amazon EventBridge for automations
    - Scan Types
        - Agent-based: Uses SSM agent to continuously scan instances
        - Agentless: Scans EBS snapshots to collect software inventory from instances
    - EC2: Uses SSM agent to scan for vulnerabilities
    - ECR: Image scanning method where it scans container images for vulnerabilities
    - Lambda: Security vulnerability assessment of lambda functions
- Amazon GuardDuty
    - Threat detection service (IDS) that uses machine learning to monitor for malicious behavior
    - Not an intrusion prevention service (IPS)
    - No installation, free 30 day trial, alerts appear in console and EventBridge, updates its own database
    - Data Sources: CloudTrail logs, VPC Flow Logs, Route53 resolver DNS query logs
    - Extended Sources: RDS/Aurora, S3, Lambda and EKS
- Amazon Macie
    - Personally Identifiable Information (PII): Home address, e-mail addresses, social security number, passport numbers, DOB, phone number, bank account, credit card numbers
    - Macie is a data security and privacy service that uses machine learning and pattern matching to discover sensitive data stored in S3 buckets
    - Can alert you via SNS and works for EventBridge
    - Great for frameworks like HIPAA, PCI, and GDPR
    - Only works for S3
- AWS Security Hub
    - Single place to view all security alerts from services like GuardDuty, Inspector, Macie, AWS Firewall Manager
    - Aggregate across multiple accounts in an organization via a delegated administrator member account
    - Single place to view all security alerts across multiple AWS security services and accounts
- AWS Audit Manager
    - Allows you to continually audit your AWS usage to make sure you stay complaint

    - Automated service that produces reports for PCI compliance, GDPR, and others
    - Works with custom assessments and frameworks
- AWS Artifact
    - Provides on-demand downloads of AWS security and compliance documents
    - Provides documents showing that services meet compliance for International Organization for Standardization (ISO) standards, Payment Card Industry (PCI) Security Standards, and System and Organization Controls (SOC)
    - Can also download compliance documents for independent software vendors (ISVs)

Infrastructure and Application Security

- AWS Certificate Manager (ACM)
    - Service that allows you to create, manage, and deploy public and private TLS certificates
    - Offers private CA option
    - Integrates with Elastic load balancing, CloudFront distributions, API Gateway
        - Do not directly associate with EC2 instances
- AWS Public TLS Certs
    - Requesting and using public TLS certificates are free
    - ACM issued TLS certificates are automatically renewed (60 days before expiration)
    - Types of Domains to know:
        - Fully Qualified Domain Name (FQDN): Complete address of a host on the internet to specify the exact location (www.google.com)
        - Wildcard Domain (*): Uses an asterisk as a subdomain to match all possible subdomains with one certificate (1 cert that works with blog.mywebsite.com and app.mywebsite.com)
    - Validation Domain Ownership
        - DNS Validation: Recommended, required for automatic renewal, create a CNAME in your DNS zone to validate
        - Email Validation: ACM sends validation emails, not recommended
    - Importing TLS Certificates
        - Import certificates that you obtained outside of AWS
        - No automatic managed renewal for imported certs
        - Notifies 45 days before expiration
    - AWS Config can use acm-certificate-expiration-check managed rule to check for expiring certs
    - Regional resources need certs deployed in same region, global resources need cert deployed in us-east-1 region
- AWS Key Manage Service (AWS KMS)
    - Managed service for creating and controlling encryption keys, integrated with AWS services, encrypts data at rest
    - Can track KMS key usage via CloudTrails and control access via IAM and key policies

- AWS KMS Keys (Used to be called Customer Master Keys (CMK))
    - Regional keys that you create and manage
    - Key Material Origin
        - AWS KMS: AWS creates the key for you
        - Custom Key Store: Key material is generated and used in an AWS CloudHSM cluster
        - Imported: Import key material from your own infrastructure and associate with key
    - Key Types
        - Customer Managed: You have full control (maintain policies, enable and disable them, automatic rotation or manual rotation, schedule deletions)
            - Never free
        - AWS Owned: Free, AWS controls everything
        - AWS Managed: Legacy, replaced by AWS Owned, KMS keys in your account that are created and used on your behalf, since it's on your account you can view and audit, but can't change anything on it
    - KMS Keys NEVER leave KMS service
- AWS KMS Key Policies
    - Primary access for controlling access to key, ONLY 1 policy per key
    - Works with IAM policies and key grants
        - Grants: Temporarily access for key use
    - Key policies must explicitly allow access to keys
- AWS KMS Multi-region Keys
    - KMS Keys are Regional
    - KMS allows multi-Region keys by creating keys in each region using same key material and key ID (still regional)
    - Perfect for consistency and redundancy for encryption across Regions
- AWS CloudHSM
    - Hardware security module (HSMs): Computing device that processes cryptographic operations and provides secure storage for cryptographic keys
    - AWS CloudHSM: AWS-based HSM enables generation and usage of your own encryption keys
    - This is a physical device, entirely dedicated to you, you have complete control, low-latency and super secure
    - Perfect for strictest requirements for cryptographic keys
- AWS Secrets Manager
    - Service that stores, encrypts, and rotates secrets and credentials
    - Encryption in transit via TLS and encryption at rest using KMS Keys
    - Automatic credential rotation for RDS and Aurora is possible
    - Never free, consider Parameter Store for cost concerns
    - Uses Lambda functions to generate new secrets when performing rotations (automatically updates credentials for you when rotating)
    - Muti-Region Secrets: Secrets data/metadata are replicated across Regions, can promote a replica secret to a standalone secret

Network Security

- AWS Shield and Shield Advanced
  - Shield Standard: Free DDoS detection and mitigation service that protects at the perimeter
    - SYN/UDP floods, reflection attacks, and other Layer 3 and 4 attacks
    - Works with Route 53 hosted zones, CloudFront distributions, Global Accelerator standard accelerators
  - Shield Advanced: Enhanced protection on even more resources, including Layer 7 traffic
    - Works on ELB, EC2, Elastic IP address, and Shield standard resources
    - 24/7 access to AS Shield Response Team(SRT)
    - Cost protection on successful DDoS attacks
    - $3,000 per month, per organization
- AWS Web Application Firewall (WAF)
  - Web Application firewall that lets you monitor HTTP and HTTPS
  - Layer 7 (Application layer) traffic
  - Supported: ALB, API Gateway, CloudFront, AppSync, Cognito User Pools
  - You define Web ACLS and Web ACL rules (can be added to rule group for reuse)
  - Rule actions: Allow, Block, Count, CAPTCHA
  - Common WAF Match Rule Statements: IP set, Geographic, Regex, SQL injection, String match, Cross-site scripting (XSS)
  - Can use a rate-based rule statement if requests are coming in too fast
- AWS Network Firewall
  - Stateful, managed, network firewall and intrusion detection and prevention service
  - Includes firewall rules engine
  - Ex: Can block outbound Server Message Block (SMB) requests
  - Protects: Layer 3, 4, and 7 in any direction
  - Inspect and filter based on IP/port, protocol, domains, and regex patterns
  - Can send logs to CloudWatch, S3, and Data Firehose
  - You can drop, allow, or alert
  - You deploy firewall endpoints and route traffic through them
- AWS Firewall Manager
  - Centrally manage multiple AWS WAF Web ACL rules across multiple accounts in an AWS Organization in a single pane of glass
  - Can manage: AWS WAF Rules, AWS Shield Advanced, VPC Security Groups, AWS Network Firewall, Route 53 Resolver DNS Firewall (All are regional resources)
  - Existing rules are automatically applied to new resources/accounts

Cost and Budgeting Services

- AWS Cost Explorer
  - Allows you to visualize, analyze, and manage cloud costs

    - Can generate custom reports based on factors like resource tags, resources, service categories, built-in forecasting
    - Recommends suggested Savings Plans
    - Cost Allocation Tags
        - AWS-generated: Applied for you by AWS (aws:createdBy)
        - User-defined: You create them
- AWS Budgets
    - Allows planning and setting expectations around cloud costs
    - Track ongoing spend and create alerts
    - Can use Cost Explorer in conjunction to create fine-grained budgets
    - Budget Actions: Run actions when a budget exceeds certain threshold
        - These can run automatically or run after manual approvals
- AWS Cost and Usage Reports (CUR/CUR2.0)
    - Beginning to transition to AWS Data Exports
    - Most comprehensive set of cost and usage data available
    - Can publish recurring billing reports to S3
    - Can be used within organization level, OU level, or member account level
- AWS Cost Anomaly Detection
    - Uses machine learning to detect and alert on unusual spend patterns, sends alerts to SNS, Slack, Amazon Chime
    - Rank anomalies based on dollar impact
    - Helps perform root cause analysis
    - Learns historic patterns to help lower false alerts
    - Split by service, account, Region, or usage type
    - Can aggregate alerts or send individual ones
- AWS License Manager
    - AWS service to simplify managing software licenses with different vendors (Microsoft, SAP, and Oracle)
    - Helps centrally manage licenses across AWS accounts and on-prem environments
    - Good for hybrid environment management and preventing license overuse